

**REPUBLIC OF RWANDA
MINISTRY OF EDUCATION**



National Examination and School Inspection Authority



**NESA END OF YEAR COMPREHENSIVE
ASSESSMENT EXAMINATIONS OF COMPUTER
SCIENCE FOR SENIOR 4 AND SENIOR 5**

SYSTEMATIZED BY COMPUTER SCIENCE DEPARTMENT

COMPUTER SCIENCE

Date: 13/ 06/2025
Period: 8:30 AM -11:30 AM



END OF TERM III EXAMINATIONS QUESTION PAPER

GRADE:

SENIOR FOUR (S4)

LEVEL:

ADVANCED LEVEL

COMBINATIONS:

- MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (**MCE**)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (**MPC**)

DURATION:

3HOURS

MARKS:

...../ 100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all **12** questions. **(55marks)**

Section B: Attempt **3** questions. **(30marks)**

Section C: Attempt **1** question. **(15marks)**

2)Use **a blue or black ink pen only** to write your answers.

Section A: Attempt all 12 questions from this section (55marks)

For multiple choice questions circle correct Answer

1) i) What is a computer? **(2 marks)**

- a) A human brain
- b) An electronic device for processing data
- c) A mechanical tool
- d) A handheld calculator

ii) Match the type of computer with its function: **(6 marks)**

Computer	Functions
a) Supercomputer	1. Control specific tasks in devices
b) Mainframe	2. Perform extensive calculations in research
c) Embedded computer	3. Process large-scale organizational data

2) What is the function of the control unit in the CPU? **(2 marks)**

- a) Performs arithmetic calculations
- b) Coordinates all processing activities
- c) Stores temporary data
- d) Handles data input

3) Match the port with its primary use: **(6 marks)**

port	Uses
a) Audio port	1. Connects computers to networks
b) Network port (RJ45)	2. Transfers data between devices
c) USB port	3. Connects speakers or microphones

4) Answer the following questions using **True/False:** **(4 marks)**

- a) Ethics in computer usage only applies to software developers, not end-users.
- b) Avoiding the use of weak passwords is a good practice for computer safety.
- c) E-waste should be taken to certified recycling centers to ensure proper handling.
- d) Throwing e-waste into regular trash bins is safe and environmentally friendly.

5) i) Convert the binary number 1010 to decimal: **(2 marks)**

- a) 5
- b) 8
- c) 10
- d) 12

ii) Convert the hexadecimal number 2F to binary. **(3 marks)**

- a) 00111111₂.
- b) 00101111₂.
- c) 01101111₂.
- d) 00101011₂.

6) What is the primary purpose of using pseudocode? **(2 marks)**

- a) To write code directly in machine language
- b) To describe algorithms in a human-readable format
- c) To design hardware components
- d) To avoid debugging

7) What is the main role of a compiler? **(3 marks)**

- a) To execute programs directly
- b) To translate source code into machine code
- c) To debug logical errors
- d) To optimize program performance

8) Match the C++ concept with its description: **(6 marks)**

- | | |
|--------------|--|
| a) Variable | 1. A container to store data |
| b) Data type | 2. Specifies the type of data a variable can hold |
| c) Function | 3. A block of code designed to perform a specific task |

9) Answer the following questions using **True/False**: **(4 marks)**

- a) An arithmetic expression in C++ evaluates from left to right, regardless of operator precedence.
- b) The expression $(5 + 2) * 3$ evaluates to 21 in C++.

10) Match the following components of an operating system with their roles: **(6 marks)**

- | | |
|----------------|---|
| a) Kernel | 1) Provides user interface to interact with the OS. |
| b) File System | 2) Manages storage and access to data. |
| c) Shell | 3) Central part managing CPU and memory. |

11) i) In HTML, what does the attribute href specify in an anchor tag? **(3 marks)**

- a) The text color
- b) The hyperlink reference
- c) The font style
- d) The button type

ii) Which tag is used to describe the content and properties of a web page but is not displayed directly on the page in an HTML document? **(2 marks)**

- a) <meta>
- b) <head>
- c) <body>
- d) <title>

12) Answer the following questions using **True/False**: **(4 marks)**

- a) The <table> element can include rows and cells to display structured data.
- b) HTML5 introduces new elements like <article>, <aside>, and <nav> to improve web semantics.

Section B: Attempt any (3) three questions from this Section (30marks)

- 13) The steps for backing up data using the Windows Backup utility are listed below in a disordered manner. Analyze and arrange them in the correct order:

(10 marks)

- A. Save the settings to enable automatic backups.
- B. Open the "Settings" menu and select "Update & Security."
- C. Click "Backup" and choose "Add a drive."
- D. Click "More options" to configure backup frequency and files.
- E. Select the drive to store the backup.

- 14) i) Which of the following logic gates can be used to implement any digital circuit?

(3 marks)

- a) AND and OR gates
- b) NAND and NOR gates
- c) XOR gates
- d) NOT gates

- ii) Critique the use of NAND gates as universal gates in modern circuits.

(7 marks)

- 15) Design a pseudocode to calculate the factorial of a given number

(10 marks)

- 16) Demonstrate the use of a while loop to find the sum of digits of an integer number entered by the user.

(10 marks)

- 17) i) Evaluate the impact of using the float property in web page layouts

(6 marks)

- ii) Evaluate the importance of using a consistent colour scheme in web design.

(4 marks)

Section C: Attempt any (1) one question from this Section (15marks)

- 18) Draw a flowchart to reads two values, determines the largest value and prints the largest value with an identifying message **(15marks)**
- 19) Design a C++ program that demonstrates the use of function overloading for calculating the area of a circle and a rectangle. **(15marks)**
- 20) Demonstrate how to reverse the elements of an array using a loop. **(15marks)**

COMPUTER SCIENCE

Date: 13/ 06/2025
Period: 8:30 AM -11:30 AM



END OF TERM III EXAMINATIONS
MARKING GUIDE

GRADE: SENIOR FOUR (S4)

LEVEL: ADVANCED LEVEL

COMBINATIONS:
-MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS **(MCE)**
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE **(MPC)**

DURATION: 3HOURS

MARKS:

...../100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all **12** questions. **(55marks)**

Section B: Attempt **3** questions. **(30marks)**

Section C: Attempt **1** question. **(15marks)**

2)Use **a blue or black ink pen only** to write your answers.

Section A: Attempt all 12 questions from this section

(55marks)

For multiple choice questions circle correct Answer

1) i) What is a computer?

(2 marks)

- a) A human brain
- b) An electronic device for processing data
- c) A mechanical tool
- d) A handheld calculator

Answer

- b) An electronic device for processing data

ii) Match the type of computer with its function:

(6 marks)

Computer

Functions

- | | |
|----------------------|---|
| a) Supercomputer | 1. Control specific tasks in devices |
| b) Mainframe | 2. Perform extensive calculations in research |
| c) Embedded computer | 3. Process large-scale organizational data |

Answer

a → 2

b → 3

c → 1

2) What is the function of the control unit in the CPU?

(2 marks)

- a) Performs arithmetic calculations
- b) Coordinates all processing activities
- c) Stores temporary data
- d) Handles data input

Answer

b) Coordinates all processing activities

3) Match the port with its primary use: **(6 marks)**

port	Uses
a) Audio port	1. Connects computers to networks
b) Network port (RJ45)	2. Transfers data between devices
c) USB port	3. Connects speakers or microphones

Answer

a → 3

b → 1

c → 2

4) Answer the following questions using **True/False:** **(4 marks)**

- a) Ethics in computer usage only applies to software developers, not end-users.
- b) Avoiding the use of weak passwords is a good practice for computer safety.
- c) E-waste should be taken to certified recycling centers to ensure proper handling.
- d) Throwing e-waste into regular trash bins is safe and environmentally friendly.

Answer

- a) False
- b) True
- c) True
- d) False

5) i) Convert the binary number 1010 to decimal: **(2 marks)**

- a) 5
- b) 8
- c) 10
- d) 12

Answer

c) 10

ii) Convert the hexadecimal number 2F to binary. **(3 marks)**

- a) 00111111₂.
- b) 00101111₂.
- c) 01101111₂.
- d) 00101011₂.

Answer

b) 00101111₂.

6) What is the primary purpose of using pseudocode? **(2 marks)**

- a) To write code directly in machine language
- b) To describe algorithms in a human-readable format
- c) To design hardware components
- d) To avoid debugging

Answer

b) To describe algorithms in a human-readable format

7) What is the main role of a compiler? **(3 marks)**

- a) To execute programs directly
- b) To translate source code into machine code
- c) To debug logical errors
- d) To optimize program performance

Answer

b) To translate source code into machine code

8) Match the C++ concept with its description: **(6 marks)**

- | | |
|--------------|--|
| a) Variable | 1. A container to store data |
| b) Data type | 2. Specifies the type of data a variable can hold |
| c) Function | 3. A block of code designed to perform a specific task |

Answer

a → 1

b → 2

c → 3

9) Answer the following questions using **True/False**: **(4 marks)**

- a) An arithmetic expression in C++ evaluates from left to right, regardless of operator precedence.
- b) The expression $(5 + 2) * 3$ evaluates to 21 in C++.

Answer

- a) False
- b) True

- 10) Match the following components of an operating system with their roles: **(6 marks)**

- | | |
|----------------|---|
| a) Kernel | 1) Provides user interface to interact with the OS. |
| b) File System | 2) Manages storage and access to data. |
| c) Shell | 3) Central part managing CPU and memory. |

Answer

A - 3, B - 2, C - 1

- 11) i) In HTML, what does the attribute href specify in an anchor tag? **(3 marks)**

- a) The text color
- b) The hyperlink reference
- c) The font style
- d) The button type

Answer

- b) The hyperlink reference

- ii) Which tag is used to describe the content and properties of a web page but is not displayed directly on the page in an HTML document? (2 marks)

- a) <meta>
- b) <head>
- c) <body>
- d) <title>

Answer

- a) <meta>

- 12) Answer the following questions using **True/False**: **(4 marks)**

- a) The <table> element can include rows and cells to display structured data.
- b) HTML5 introduces new elements like <article>, <aside>, and <nav> to improve web semantics.

Answer

- a) True
- b) True

Section B: Attempt any (3) three questions from this Section (30marks)

- 13) The steps for backing up data using the Windows Backup utility are listed below in a disordered manner. Analyze and arrange them in the correct order:

(10 marks)

- A. Save the settings to enable automatic backups.
- B. Open the "Settings" menu and select "Update & Security."
- C. Click "Backup" and choose "Add a drive."
- D. Click "More options" to configure backup frequency and files.
- E. Select the drive to store the backup.

Answer

Correct Order:

- 1. B. Open the "Settings" menu and select "Update & Security."
 - 2. C. Click "Backup" and choose "Add a drive."
 - 3. E. Select the drive to store the backup.
 - 4. D. Click "More options" to configure backup frequency and files.
 - 5. A. Save the settings to enable automatic backups.
- 14) i) Which of the following logic gates can be used to implement any digital circuit? **(3 marks)**

- a) AND and OR gates
- b) NAND and NOR gates
- c) XOR gates
- d) NOT gates

Answer

- b) NAND and NOR gates

- ii) Critique the use of NAND gates as universal gates in modern circuits.

(7 marks)

Answer

Advantages:

- ✓ Simplifies design by reducing the number of gate types needed.
- ✓ Universal nature allows implementation of any Boolean function.
- ✓ Reduces manufacturing cost and complexity in IC design.

Challenges:

- ✓ Increased power consumption in large-scale circuits.
- ✓ More gates may be required for complex logic.
- ✓ Higher delay due to additional gate levels.

- 15) Design a pseudocode to calculate the factorial of a given number
(10 marks)

Answer

```
Start
Input n
Factorial = 1
For i = 1 to n:
    Factorial = Factorial × i
i=i+1
End For
Display Factorial
Stop
```

- 16) Demonstrate the use of a while loop to find the sum of digits of an integer number entered by the user.
(10 marks)

Answer

```
#include<iostream>
using namespace std;
int main() {
    int num, sum = 0;
    cout << "Enter an integer number: ";
    cin >> num;
    while (num > 0) {
        sum += num % 10;
        num /= 10;
    }
    cout << "Sum of digits: " << sum << endl;
    return 0;
}
```

- 17) i) Evaluate the impact of using the float property in web page layouts
(6 marks)

Answer

- Allows elements to be positioned left or right.
- Enables text or elements to wrap around floated items.
- Can cause layout issues if clear properties are not used.

- ii) Evaluate the importance of using a consistent color scheme in web design.
(4 marks)

Answer

- Enhances visual appeal.
- Creates a cohesive user experience.
- Improves readability.

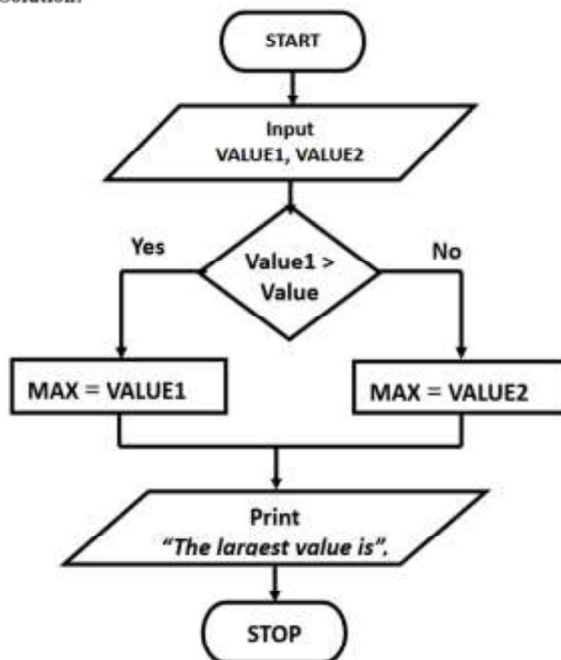
- Reinforces brand identity.
- Increases accessibility.
- Reduces user confusion.

Section C: Attempt any (1) one question from this Section (15marks)

- 18) Draw a flowchart to reads two values, determines the largest value and prints the largest value with an identifying message **(15marks)**

Answer

Solution:



- 19) Design a C++ program that demonstrates the use of function overloading for calculating the area of a circle and a rectangle. **(15marks)**

Answer

```

#include <iostream>
using namespace std;
double area(double radius) {
return 3.14159 * radius * radius;
}
double area(double length, double breadth) {
return length * breadth;
}
int main() {
double radius, length, breadth;

```

```

cout << "Enter radius of the circle: ";
cin >> radius;
cout << "Area of circle: " << area(radius) << endl;

cout << "Enter length and breadth of the rectangle: ";
cin >> length >> breadth;
cout << "Area of rectangle: " << area(length, breadth) << endl;
return 0;
}

```

20) Demonstrate how to reverse the elements of an array using a loop.

(15marks)

Answer

```

#include <iostream>
using namespace std;
int main() {
int arr[5], temp;
cout << "Enter 5 elements: \n";
for (int i = 0; i < 5; i++) {
cin >> arr[i];
}
for (int i = 0; i < 5 / 2; i++) {
temp = arr[i]; arr[i] = arr[4 - i]; arr[4 - i] = temp;
}
cout << "Reversed array: \n";
for (int i = 0; i < 5; i++) {

cout << arr[i] << " ";
}
return 0;
}

```

COMPUTER SCIENCE

Date: 13/ 06/2025
Period: 8:30 AM -11:30 AM



END OF TERM III EXAMINATIONS QUESTION PAPER

GRADE: SENIOR FIVE (S5)

LEVEL: ADVANCED LEVEL

COMBINATIONS: -MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (MCE)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (MPC)

DURATION: 3HOURS

MARKS:

...../100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all **12** questions. (55marks)

Section B: Attempt **3** questions. (30marks)

Section C: Attempt **1** question. (15marks)

2)Use a **blue or black ink pen only** to write your answers.

Section A: Attempt all 12 questions from this section

(55marks)

1) Match the maintenance tip with the corresponding laptop part. **(4 marks)**

- | | |
|-------------|---|
| a) Screen | 1) Use compressed air to remove dust. |
| b) Keyboard | 2) Clean with an approved LCD cleaner and a soft cloth. |
| c) Battery | 3) Avoid overcharging and store in a cool, dry place. |
| d) Ports | 4) Ensure all connections are secure and free of dust. |

2) i) Which type of transmission medium uses copper wires for data transmission? **(2 marks)**

- a) Twisted-pair cable
- b) Coaxial cable
- c) Fiber optic cable
- d) Radio waves

ii) What is the primary function of a computer network? **(2 marks)**

- a) To connect computers and devices for sharing resources and communication.
- b) To store and process data.
- c) To run applications.
- d) To control hardware.

3) i) What is the primary purpose of a database? **(2 marks)**

- a) To store and manage data efficiently.
- b) To perform complex calculations.
- c) To display information on a website.
- d) To send emails.

ii) Why is data redundancy a problem in traditional file systems? **(3 marks)**

- a) It increases data integrity.
- b) It leads to storage wastage and inconsistencies.
- c) It enhances data sharing among users.
- d) It ensures data security.

4) Why are referential columns used in relational databases? **(3 marks)**

- a) To encrypt data stored in tables
- b) To link related data between tables
- c) To increase the size of database tables
- d) To simplify user queries

5) What does encapsulation refer to in C++? **(3 marks)**

- a) Binding data and functions together in a class
- b) Hiding class methods
- c) Defining multiple functions with the same name
- d) Allocating dynamic memory

6) Analyze the following statements about operator overloading and its effect on program readability. For each statement, determine whether it is **True** or **False**. **(6 marks)**

- a) Operator overloading allows the same operator to be used for different data types, making code more intuitive and readable.
- b) Using operator overloading excessively can make the code harder to understand, as it may confuse the reader about the operator's intended functionality.
- c) Operator overloading can reduce the readability of code when it is applied to complex types that the reader may not be familiar with.

7) i) Which version of Visual Basic introduced Rapid Application Development (RAD)? **(2 marks)**

- a) Visual Basic 4.0
- b) Visual Basic 5.0
- c) Visual Basic 6.0
- d) Visual Basic .NET

iii) What is the difference between an Option Button and a CheckBox in Visual Basic? **(3 marks)**

- a) Option Buttons allow multiple selections, while CheckBoxes do not.
- b) Option Buttons are used for text input, while CheckBoxes are not.
- c) Option Buttons allow single selection, while CheckBoxes allow multiple selections.
- d) Both serve the same purpose.

8) i) Which data type is used to store a large range of numeric values in Visual Basic? **(3 marks)**

- a) Byte
- b) Integer
- c) Long
- d) Boolean

ii) Which of the following is a valid variable name in Visual Basic? **(3 marks)**

- a) 2Years
- b) My.house
- c) First_Name
- d) Her&hermother

9) Which logical operator in Visual Basic evaluates to True only when both operands are False? **(3 marks)**

- a) Or
- b) Not
- c) And
- d) Xor

10) i) Which command is used to compile a Java source code file? **(2 marks)**

- a) javac
- b) java
- c) javaw
- d) javap

ii) Which of the following is not a Java primitive data type? **(2 marks)**

- a) Boolean
- b) Float
- c) String
- d) Byte

11) Match the Java IO components with their roles: **(6 marks)**

- | | |
|---------------|---------------------------------|
| a) System.in | 1) Input data from the keyboard |
| b) System.out | 2) Output error messages |
| c) System.err | 3) Output normal results |

12) Represent the following employee relation in the conceptual model

(6 marks)

Employees relation				
Empld	FirstName	Surname	Address	Birthdate

Section B: Attempt any (3) three questions from this Section (30 marks)

13) You are tasked with designing a network for a small office with 20 computers, 3 printers, and a server. Describe how you would implement the chosen topology in the physical office space. Consider cable routing, device placement, and equipment configuration. **(10 marks)**

14) Write a C++ program demonstrating multiple inheritance **(10 marks)**

15) Develop a VB 6.0 or VB. Net application that display the following pattern of numbers on the Label. **(10 marks)**



```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

16) Design a Java program to calculate and print the factorial of a given number. **(10 marks)**

17) Create a Java program that defines a Student class with attributes name and age. Implement a parameterised constructor to initialize these attributes, and a displayInfo() method to display the student's information. **(10 marks)**

Section C: Attempt any (1) one question from this Section (15 marks)

18) Write a C++ program that uses bubble sort technic to arrange 10 elements entered by user into an array in ascending order and display the smallest element among them. **(15 marks)**

18) Analyze the advantages and risks of using dynamic memory allocation in C++. **(15 marks)**

20) Create a program to demonstrate the use of super for calling a parent class constructor. **(15 marks)**

COMPUTER SCIENCE

Date: 13/ 06/2025
Period: 8:30 AM -11:30 AM



END OF TERM III EXAMINATIONS MARKING GUIDE

GRADE: **SENIOR FIVE (S5)**

LEVEL: **ADVANCED LEVEL**

COMBINATIONS: -MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (**MCE**)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (**MPC**)

DURATION: **3HOURS**

MARKS:

...../100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all **12** questions. **(55marks)**

Section B: Attempt **3** questions. **(30marks)**

Section C: Attempt **1** question. **(15marks)**

2)Use **a blue or black ink pen only** to write your answers.

Section A: Attempt all 12 questions from this section

(55marks)

1) Match the maintenance tip with the corresponding laptop part. **(4 marks)**

- | | |
|-------------|---|
| a) Screen | 1) Use compressed air to remove dust. |
| b) Keyboard | 2) Clean with an approved LCD cleaner and a soft cloth. |
| c) Battery | 3) Avoid overcharging and store in a cool, dry place. |
| d) Ports | 4) Ensure all connections are secure and free of dust. |

Answer

a-2, b-1, c-3 d-4

2) i) Which type of transmission medium uses copper wires for data transmission? **(2 marks)**

- a) Twisted-pair cable
- b) Coaxial cable
- c) Fiber optic cable
- d) Radio waves

Answer

- a) Twisted-pair cable

ii) What is the primary function of a computer network? **(2 marks)**

- a) To connect computers and devices for sharing resources and communication.
- b) To store and process data.
- c) To run applications.
- d) To control hardware.

Answer

a) To connect computers and devices for sharing resources and communication.

3) i) What is the primary purpose of a database? **(2 marks)**

- a) To store and manage data efficiently.

- b) To perform complex calculations.
- c) To display information on a website.
- d) To send emails.

Answer

- a) To store and manage data efficiently.
- ii) Why is data redundancy a problem in traditional file systems? **(3 marks)**
- a) It increases data integrity.
 - b) It leads to storage wastage and inconsistencies.
 - c) It enhances data sharing among users.
 - d) It ensures data security.

Answer

- b) It leads to storage wastage and inconsistencies.
- 4) Why are referential columns used in relational databases? **(3 marks)**
- a) To encrypt data stored in tables
 - b) To link related data between tables
 - c) To increase the size of database tables
 - d) To simplify user queries

Answer

- b) To link related data between tables
- 5) What does encapsulation refer to in C++? **(3 marks)**
- a) Binding data and functions together in a class
 - b) Hiding class methods
 - c) Defining multiple functions with the same name
 - d) Allocating dynamic memory

Answer

- a) Binding data and functions together in a class
- 6) Analyze the following statements about operator overloading and its effect on program readability. For each statement, determine whether it is True or False. **(6 marks)**
- a) Operator overloading allows the same operator to be used for different data types, making code more intuitive and readable.

- b) Using operator overloading excessively can make the code harder to understand, as it may confuse the reader about the operator's intended functionality.
- c) Operator overloading can reduce the readability of code when it is applied to complex types that the reader may not be familiar with.

Answer

- a) True
 - b) True
 - c) True
- 7) i) Which version of Visual Basic introduced Rapid Application Development (RAD)? **(2 marks)**

- a) Visual Basic 4.0
- b) Visual Basic 5.0
- c) Visual Basic 6.0
- d) Visual Basic .NET

Answer

- c) Visual Basic 6.0
- iii) What is the difference between an Option Button and a CheckBox in Visual Basic? **(3 marks)**

- a) Option Buttons allow multiple selections, while CheckBoxes do not.
- b) Option Buttons are used for text input, while CheckBoxes are not.
- c) Option Buttons allow single selection, while CheckBoxes allow multiple selections.
- d) Both serve the same purpose.

Answer

- c) Option Buttons allow single selection, while CheckBoxes allow multiple selections.
- 8) i) Which data type is used to store a large range of numeric values in Visual Basic? **(3 marks)**

- a) Byte
- b) Integer
- c) Long

d) Boolean

Answer

c) Long

ii) Which of the following is a valid variable name in Visual Basic? **(3 marks)**

a) 2Years

b) My.house

c) First_Name

d) Her&hermother

Answer

c) First_Name

9) Which logical operator in Visual Basic evaluates to True only when both operands are False? **(3 marks)**

a) Or

b) Not

c) And

d) Xor

Answer

d) Xor

10) i) Which command is used to compile a Java source code file? **(2 marks)**

a) javac

b) java

c) javaw

d) javap

Answer

a) javac

ii) Which of the following is not a Java primitive data type? **(2 marks)**

a) Boolean

b) Float

c) String

d) Byte

Answer

c) String

11) Match the Java IO components with their roles: **(6 marks)**

- | | |
|---------------|---------------------------------|
| a) System.in | 1) Input data from the keyboard |
| b) System.out | 2) Output error messages |
| c) System.err | 3) Output normal results |

Answer

a - 1, b - 3, C - 2

12) Represent the following employee relation in the conceptual model

(6 marks)

Employees relation				
Empld	FirstName	Surname	Address	Birthdate

Answer



Section B: Attempt any (3) three questions from this Section (30 marks)

13) You are tasked with designing a network for a small office with 20 computers, 3 printers, and a server. Describe how you would implement the chosen topology in the physical office space. Consider cable routing, device placement, and equipment configuration. **(10 marks)**

Answer

Implementation steps:

- Install a central switch in a secure and accessible location.
- Run network cables (Cat5e or Cat6) from the switch to each device, following established standards and avoiding clutter.
- Configure the switch to assign IP addresses and manage network traffic.

- Secure the network with firewalls and access control measures on the switch and devices.

14) Write a C++ program demonstrating multiple inheritance **(10 marks)**

Answer

```
#include <iostream>

using namespace std;

class Person {

    public:

    void displayPerson() {

        cout << "This is a Person." << endl;

    } };

class Employee {

    public:

    void displayEmployee() {

        cout << "This is an Employee." << endl;

    } };

class Manager : public Person, public Employee {};

int main() { Manager mgr;

    mgr.displayPerson();

    mgr.displayEmployee();

    return 0;}
```

15) Develop a VB 6.0 or VB. Net application that display the following pattern of numbers on the Label. **(10 marks)**



Answer

VB6.0	VB.net
<pre> Private Sub Form_Activate() Dim x As Integer x = 5 Label1.Caption = "" For i = 1 To x For j = 1 To i Label1.Caption = Label1.Caption & " " & j Next j Label1.Caption = Label1.Caption & vbCrLf Next i For i = x - 1 To 1 Step -1 For j = 1 To i </pre>	<pre> Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load Dim x As Integer x = 5 For i = 1 To x For j = 1 To i Label1.text &= j & " " Next j Label1.Text &= vbCrLf Next i For i = x - 1 To 1 Step -1 For j = 1 To i Label1.Text &= j & " " </pre>

Label1.Caption = Label1.Caption & " " & j Next j Label1.Caption = Label1.Caption & vbCrLf Next i End Sub	Next j Label1.Text &= vbCrLf Next i End Sub End Class
--	---

- 16) Design a Java program to calculate and print the factorial of a given number. **(10 marks)**

Answer

```
public class Factorial {
    public static void main(String[] args) {
        int num = 5;
        int factorial = 1;
        for (int i = 1; i <= num; i++) {
            factorial *= i; }
        System.out.println("Factorial of " + num + " is: " + factorial);
    }
}
```

- 17) Create a Java program that defines a Student class with attributes name and age. Implement a parameterised constructor to initialize these attributes, and a displayInfo() method to display the student's information.

(10 marks)

Answer

```
public class Student {
    private String name;
    private int age;
    public Student(String name, int age) {
```

```
        this.name = name;
        this.age = age;
    }
    public void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }

    public static void main(String[] args) {
        Student student1 = new Student("Alice", 20);
        Student student2 = new Student("Bob", 22);
        student1.displayInfo();
        student2.displayInfo();
    } }
```

Section C: Attempt any (1) one question from this Section (15 marks)

18) Write a C++ program that uses bubble sort technic to arrange 10 elements entered by user into an array in ascending order and display the smallest element among them. **(15 marks)**

Answer

```
#include<iostream>
using namespace std;
int main(){
    int arr[10];
    int n=10,i,j,temp;
    cout<<"Enter 10 integer elements"<<endl;
    for(i=0;i<10;i++){
        cin>>arr[i];
    }
    cout<<"The List of elements to be sorted are: "<<endl;

    for(i=0;i<n;i++){
        cout<<arr[i]<<" ";
    }
    cout<<endl;
    for(i=0;i<n-1;i++){
        for(j=0;j<n-i-1;j++){
            if(arr[j]>arr[j+1]){
                temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp; }
        }
    }

    cout<<"After sorting out your list of elements, the smallest is:"<<arr[0]<<endl;

    return 0;
}
```

18) Analyze the advantages and risks of using dynamic memory allocation in C++. **(15 marks)**

Answer

Advantages:

1. Memory can be allocated as needed during runtime.
2. Enables creation of dynamic data structures like linked lists.
3. Efficient use of memory for variable-sized data.
4. Reduces waste in memory allocation.
5. Allows implementation of flexible algorithms.

Risks:

1. Risk of memory leaks if not deallocated properly.
2. Potential segmentation faults if accessing invalid memory.
3. Increased program complexity.
4. Dependency on system resources for allocation.
5. Deallocation errors may cause undefined behaviour.

20) Create a program to demonstrate the use of super for calling a parent class constructor. **(15 marks)**

Answer

```
class Parent {  
    Parent() {  
        System.out.println("Parent class constructor.");  
    } }  
  
class Child extends Parent {  
    Child() {  
        super(); // Calls parent class constructor  
        System.out.println("Child class constructor.");  
    } }  
  
public class SuperConstructor {  
    public static void main(String[] args) {  
        Child child = new Child();  
    } }
```

National Examination Paper:

Computer Science

Level: S6



MOCK FOR THE 2024/2025 NATIONAL EXAMINATIONS

QUESTION PAPER

**COMBINATIONS/
OPTION/TRADE:**

MCE and MPC

CANDIDATE NAMES:

INDEX NUMBER:

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

1. Read and understand all instructions before answering.
2. For multiple-choice questions, choose only one correct answer and clearly circle the letter or number.
3. Erasing or altering answers is strictly prohibited and will lead to zero mark for the affected question.
4. All answers for open-Ended questions must be written in the spaces provided herein.
5. Use only blue or black ink pens (no pencils or colored pens)

May 2025

1) What does a computer do during the processing phase? (2 mark)

- a) Stores data permanently
- b) Translates code into machine language
- c) Executes operations based on instructions
- d) Deletes unwanted information

2) Match the following terms with their correct descriptions: (3 marks)

Terms	Descriptions
a) Supercomputer	1) Stores data temporarily
b) ROM	2) Used for large-scale scientific computations
c) Integrated Circuits	3) Introduced in the third generation of computers

3) Which innovation could improve computer assembly processes? (2 marks)

- a) Advanced robotic arms
- b) Wireless network adapters
- c) PS/2 ports
- d) Parallel interfaces

4) Match the internal component to its function: (3 marks)

- a) Power supply unit
- b) Processor
- c) Motherboard

Functions:

- 1. Connects all components and enables communication
- 2. Supplies power to all components
- 3. Performs all processing tasks

5) Why is it unethical to use pirated software? (2 marks)

- a) It is outdated
- b) It violates intellectual property rights
- c) It has fewer features
- d) It cannot be installed on modern computers

6) Which of the following is a safe practice when using public Wi-Fi? (2 marks)

- a) Accessing sensitive accounts such as online banking

- b) Using a Virtual Private Network (VPN)
- c) Disabling firewalls for faster internet speed
- d) Turning off antivirus software

7) Which of the following represents the binary equivalent of the decimal number 13? (1 mark)

- a) 1101
- b) 1011
- c) 1110
- d) 1001

8) Determine whether the following statements about low-level and high-level programming languages are True or False: (4 marks)

- 1. Low-level languages are closer to machine code.
- 2. High-level languages are harder to debug.
- 3. Low-level languages give better hardware control.
- 4. High-level languages don't need a compiler.

9) Which of the following is a ternary operator in C++? (1 mark)

- a) ? :
- b) ::
- c) ,
- d) &

10) What is the output of the following C++ code snippet? (2 marks)

```
for (int i = 1; i <= 5; ++i) {  
    if (i == 3) break;  
    cout << i << " ";  
}
```

- a) 1 2
- b) 1 2 3
- c) 1 2 3 4 5
- d) Compilation error

11) What is the key difference between UNIX and Linux? (2 marks)

- a) Linux is proprietary
- b) UNIX supports single-user environments
- c) Linux is a UNIX-compatible open-source system
- d) UNIX lacks command-line interfaces

12) Why is migrating from HTML4 to HTML5 considered beneficial? (2 marks)

- a) HTML5 eliminates support for CSS.
- b) HTML5 supports modern multimedia elements without plugins like Flash.

- c) HTML5 removes hyperlinks.
- d) HTML5 is not compatible with older browsers.

- 13) Complete the table below by selecting the most appropriate cleaning method for each component from the options provided. (3 marks)

Laptop Component	Cleaning Procedure
1. Laptop Screen	
2. Keyboard	
3. Touchpad	

Options (choose the most appropriate for each)

- a) Avoid excessive moisture and gently wipe the surface with an alcohol-based wipe or microfiber cloth.
 - b) Use a microfiber cloth dampened with a cleaning solution.
 - c) Use compressed air to blow out dirt.
- 14) What does a node in a linked list typically contain? (2 marks)
- a) Data and a pointer to the next node
 - b) Data and its index
 - c) Only the data
 - d) Only a pointer
- 15) What is the role of the Access Point in a wireless network? (2 marks)
- a) To manage IP addresses of connected devices
 - b) To enable wired connections to switches
 - c) To act as a central device for connecting wireless devices
 - d) To prevent unauthorized access to a network
- 16) What is the goal of database normalization? (2 marks)
- a) To reduce query execution time
 - b) To eliminate redundancy and improve consistency
 - c) To enhance database encryption
 - d) To increase the size of database tables
- 17) Why are structures used in C++? (2 marks)
- a) To perform arithmetic operations
 - b) To group variables of different types under one name
 - c) To create loops
 - d) To manage pointers
- 18) Evaluate the efficiency of using structures for grouping data over separate variables. (5 marks)

.....

.....
.....
.....
.....
.....

- 19) What is the role of a destructor in C++? (2 marks)
- a) To delete private data
 - b) To release resources allocated to an object
 - c) To create copies of an object
 - d) To enable inheritance
- 20) How does a For...Next loop differ from a While loop? (2 marks)
- a) A For loop executes a fixed number of times, while a While loop runs indefinitely.
 - b) A For loop uses a counter variable, while a While loop uses a condition.
 - c) A For loop cannot include nested loops, while a While loop can.
 - d) A For loop executes faster than a While loop.
- 20) Why is Java considered platform-independent? (2 marks)
- a) It runs directly on hardware.
 - b) Java programs are compiled into bytecode that runs on any JVM.
 - c) It uses machine code instead of high-level code.
 - d) It doesn't need a compiler or interpreter.
- 21) Determine whether the following statements is True or False (2 marks)
- a) A BufferedReader can be combined with an InputStreamReader for efficient reading of text from a file.
 - b) Using FileOutputStream is a poor choice when working with text data, as it writes byte-level data.
- 22) What is the difference between default and parameterized constructors? (2 marks)
- a) Default constructors initialize objects with parameters, while parameterized constructors do not.
 - b) Parameterized constructors require arguments, while default constructors do not.
 - c) Default constructors are defined by the programmer, while parameterized constructors are provided by the compiler.
 - d) Default constructors do not exist in Java.
- 23) What does the term "stream" refer to in Java IO? (2 marks)
- a) A sequence of operations
 - b) A sequence of data values from an input source or output destination
 - c) A method to handle exceptions
 - d) A Java thread for background processes
- 24) Determine whether each statement regarding the importance of the

MAC protocol in LANs is True or False.

(2 marks)

- a) The MAC protocol in a LAN is primarily responsible for ensuring that only one device transmits data at a time to prevent collisions.
- b) The MAC protocol plays a crucial role in managing the physical layer of the OSI model by controlling how devices access the shared network medium.

25) Which type of attack involves sending repeated requests to overwhelm a system?

(2 marks)

- a) SQL Injection
- b) Trojan Horse
- c) Denial-of-Service
- d) Eavesdropping

26) Correct any mistake in this SQL query to list all employees earning more than 50,000 from a table called Employee.

SELECT WHERE * FROM Employee Salary < 50000; (2 marks)

.....
.....

27) How does a NATURAL JOIN differ from other types of joins? (2 marks)

- a) It combines all rows from both tables, regardless of matching conditions.
- b) It explicitly requires join conditions to be specified using ON or USING clauses.
- c) It automatically matches columns with the same name and eliminates duplicates.
- d) It only retrieves rows that exist in both tables without considering matching columns.

28) What does the Len function in Visual Basic return? (2 marks)

- a) The number of spaces in a string
- b) The length of a string including spaces
- c) The number of characters excluding spaces
- d) The ASCII value of a string

29) Determine whether the following statements about the key differences between DAO and ADO in Visual Basic are True or False: (2 marks)

1. DAO is for desktop apps, while ADO is for web and distributed apps.
2. ADO is more flexible, accessing multiple data sources via OLE DB.

30) What is the role of the long-term scheduler? (2 marks)

- a) Assigns CPU to processes
- b) Loads processes from secondary storage to memory

- c) Manages swapping of processes
- d) Handles I/O device allocation

31) Why is cache memory used between the CPU and main memory? (2 marks)

- a) To reduce access time for frequently used data
- b) To increase memory size
- c) To store permanent data
- d) To improve memory allocation

32) Why is ArrayList preferred over arrays in Java? (2 marks)

- a) Arrays are faster but less flexible than ArrayLists.
- b) ArrayLists resize dynamically and provide utility methods for manipulation.
- c) Arrays allow insertion at any position.
- d) ArrayLists are thread-safe by default.

33) Why are vector graphics preferred for scalable images? (2 marks)

- a) They use less memory.
- c) They do not lose quality when resized.
- d) They are easier to edit.
- b) They are pixel-based.

34) Why is the close() function essential in file handling? (2 marks)

- a) It prevents memory leaks by releasing resources used by the file.
- b) It automatically deletes the file after use.
- c) It ensures all data is delete from the file before closing.
- d) It prevents other programs from accessing the file permanently.

35)What is the significance of the JSP engine Jasper in Tomcat? (2 marks)

- a) It parses and compiles Java code
- b) It handles servlet requests
- c) It converts JSP files into servlets
- d) It provides network connections

36) Determine whether the following statements about the importance of user interface principles in project success are True or False: (2 marks)

1. A well-designed user interface improves user experience and makes the application more intuitive.
2. Focusing on aesthetic design alone is enough for a successful user interface.

37) Design a circuit that implements the Boolean expression $Z=(A \cdot \neg B)+(C \cdot D)$. Include a labelled diagram. (5 marks)

.....

38) Write a C++ program to print a multiplication table up to a given range using nested for loops. Your output must look like the following if the user input 10 as the range. (5 marks)

[illegible]

39) Write a program that demonstrates encapsulation using a class. (5 marks)

39) Write a program that demonstrates encapsulation using a class. (5 marks)

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dotted lines spaced evenly down the page, providing a guide for handwriting practice. The background is white, and there are no margins or other markings present.

40) Write a Java code snippet to demonstrate adding and retrieving elements from an ArrayList. (5 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

END

National Examination Paper:

Computer Science

Level: S6



MOCK FOR THE 2024/2025 NATIONAL EXAMINATIONS

MARKING GUIDE

COMBINATION/ OPTION/TRADE:	MPC and MCE
---------------------------------------	--------------------

May 2025

- 1) What does a computer do during the processing phase? (2 mark)
- a) Stores data permanently
 - b) Translates code into machine language
 - c) Executes operations based on instructions
 - d) Deletes unwanted information

Answer

- c) Executes operations based on instructions

- 2) Match the following terms with their correct descriptions: (3 marks)

Terms	Descriptions
a) Supercomputer	1) Stores data temporarily
b) ROM	2) Used for large-scale scientific computations
c) Integrated Circuits	3) Introduced in the third generation of computers

Answer

- a) Supercomputer → 2) Used for large-scale scientific computations
- b) ROM → 1) Stores data temporarily
- c) Integrated Circuits → 3) Introduced in the third generation of computers

- 3) Which innovation could improve computer assembly processes? (2 marks)

- a) Advanced robotic arms
- b) Wireless network adapters
- c) PS/2 ports
- d) Parallel interfaces

Answer

- a) Advanced robotic arms

- 4) Match the internal component to its function: (3 marks)

- a) Power supply unit
- b) Processor
- c) Motherboard

Functions:

- 1. Connects all components and enables communication
- 2. Supplies power to all components
- 3. Performs all processing tasks

Answer

a → 2

b → 3

c → 1

5) Why is it unethical to use pirated software? (2 marks)

- a) It is outdated
- b) It violates intellectual property rights
- c) It has fewer features
- d) It cannot be installed on modern computers

Answer

- b) It violates intellectual property rights

6) Which of the following is a safe practice when using public Wi-Fi? (2 marks)

- a) Accessing sensitive accounts such as online banking
- b) Using a Virtual Private Network (VPN)
- c) Disabling firewalls for faster internet speed
- d) Turning off antivirus software

Answer

- b) Using a Virtual Private Network (VPN)

7) Which of the following represents the binary equivalent of the decimal number 13? (1 mark)

- a) 1101
- b) 1011
- c) 1110
- d) 1001

Answer

- a) 1101

8) Determine whether the following statements about low-level and high-level programming languages are True or False: (4 marks)

1. Low-level languages are closer to machine code.
2. High-level languages are harder to debug.
3. Low-level languages give better hardware control.
4. High-level languages don't need a compiler.

Answer

1. True
2. False
3. True
4. False

9) Which of the following is a ternary operator in C++? (1 mark)

- a) ? :
- b) ::
- c) ,
- d) &

Answer

- a) ? :

10) What is the output of the following C++ code snippet? (2 marks)

```
for (int i = 1; i <= 5; ++i) {
    if (i == 3) break;
    cout << i << " ";
}
```

- a) 1 2
- b) 1 2 3
- c) 1 2 3 4 5
- d) Compilation error

Answer

- a) 1 2

11) What is the key difference between UNIX and Linux? (2 marks)

- a) Linux is proprietary
- b) UNIX supports single-user environments
- c) Linux is a UNIX-compatible open-source system
- d) UNIX lacks command-line interfaces

Answer

- c) Linux is a UNIX-compatible open-source system

12) Why is migrating from HTML4 to HTML5 considered beneficial? (2 marks)

- a) HTML5 eliminates support for CSS.
- b) HTML5 supports modern multimedia elements without plugins like Flash.
- c) HTML5 removes hyperlinks.
- d) HTML5 is not compatible with older browsers.

Answer

- b) HTML5 supports modern multimedia elements without plugins like Flash.

13) Complete the table below by selecting the most appropriate cleaning method for each component from the options provided. (3 marks)

Laptop Component	Cleaning Procedure
1. Laptop Screen	
2. Keyboard	
3. Touchpad	

Options (choose the most appropriate for each)

- a) Avoid excessive moisture and gently wipe the surface with an alcohol-based wipe or microfiber cloth.
- b) Use a microfiber cloth dampened with a cleaning solution.
- c) Use compressed air to blow out dirt.

Answer

Laptop Component	Cleaning Procedure
1. Laptop Screen	b) Use a microfiber cloth dampened with a cleaning solution.
2. Keyboard	c) Use compressed air to blow out dirt.
3. Touchpad	a) Avoid excessive moisture and gently wipe the surface with an alcohol-based wipe or microfiber cloth

- 14) What does a node in a linked list typically contain? (2 marks)
- a) Data and a pointer to the next node
 - b) Data and its index
 - c) Only the data
 - d) Only a pointer

Answer

- 15) What is the role of the Access Point in a wireless network? (2 marks)
- a) To manage IP addresses of connected devices
 - b) To enable wired connections to switches
 - c) To act as a central device for connecting wireless devices
 - d) To prevent unauthorized access to a network

Answer

- d) To act as a central device for connecting wireless devices
- 16) What is the goal of database normalization? (2 marks)
- a) To reduce query execution time
 - b) To eliminate redundancy and improve consistency

- c) To enhance database encryption
- d) To increase the size of database tables

Answer

- b) To eliminate redundancy and improve consistency
- 17) Why are structures used in C++? (2 marks)
- a) To perform arithmetic operations
 - b) To group variables of different types under one name
 - c) To create loops
 - d) To manage pointers

Answer

- b) To group variables of different types under one name
- 18) Evaluate the efficiency of using structures for grouping data over separate variables. (5 marks)

Answer

1. Simplifies management of related data.
2. Reduces code duplication.
3. Enhances readability and maintainability.
4. Enables logical grouping of data.
5. Facilitates code reusability in functions and programs.
6. Supports nested and complex data organization.
7. Improves consistency across the program.

- 19) What is the role of a destructor in C++? (2 marks)
- a) To delete private data
 - b) To release resources allocated to an object
 - c) To create copies of an object
 - d) To enable inheritance

Answer

- b) To release resources allocated to an object
- 20) How does a For...Next loop differ from a While loop? (2 marks)
- a) A For loop executes a fixed number of times, while a While loop runs indefinitely.
 - b) A For loop uses a counter variable, while a While loop uses a condition.
 - c) A For loop cannot include nested loops, while a While loop can.
 - d) A For loop executes faster than a While loop.

Answer

- b) A For loop uses a counter variable, while a While loop uses a condition.
- 20) Why is Java considered platform-independent? (2 marks)

- a) It runs directly on hardware.
- b) Java programs are compiled into bytecode that runs on any JVM.
- c) It uses machine code instead of high-level code.
- d) It doesn't need a compiler or interpreter.

Answer

b)Java programs are compiled into bytecode that runs on any JVM.

21) Determine whether the following statements is True or False
(2 marks)

- a) A BufferedReader can be combined with an InputStreamReader for efficient reading of text from a file.
- b) Using FileOutputStream is a poor choice when working with text data, as it writes byte-level data.

Answer

- a) True
- b) True

22) What is the difference between default and parameterized constructors?
(2 marks)

- a) Default constructors initialize objects with parameters, while parameterized constructors do not.
- b) Parameterized constructors require arguments, while default constructors do not.
- c) Default constructors are defined by the programmer, while parameterized constructors are provided by the compiler.
- d) Default constructors do not exist in Java.

Answer

b) Parameterized constructors require arguments, while default constructors do not.

23) What does the term "stream" refer to in Java IO? (2 marks)

- a) A sequence of operations
- b) A sequence of data values from an input source or output destination
- c) A method to handle exceptions
- d) A Java thread for background processes

Answer

b) A sequence of data values from an input source or output destination

24) Determine whether each statement regarding the importance of the MAC protocol in LANs is True or False. (2 marks)

- a) The MAC protocol in a LAN is primarily responsible for ensuring that only one device transmits data at a time to prevent collisions.

- b) The MAC protocol plays a crucial role in managing the physical layer of the OSI model by controlling how devices access the shared network medium.

Answer

- a) True
- b) True

- 25) Which type of attack involves sending repeated requests to overwhelm a system?

(2 marks)

- a) SQL Injection
- b) Trojan Horse
- c) Denial-of-Service
- d) Eavesdropping

Answer

- c) Denial-of-Service

- 26) Correct any mistake in this SQL query to list all employees earning more than 50,000 from a table called Employee.

SELECT WHERE * FROM Employee Salary < 50000; (2 marks)

Answer

SELECT * FROM Employee WHERE Salary > 50000;

- 27) How does a NATURAL JOIN differ from other types of joins? (2 marks)

- a) It combines all rows from both tables, regardless of matching conditions.
- b) It explicitly requires join conditions to be specified using ON or USING clauses.
- c) It automatically matches columns with the same name and eliminates duplicates.
- d) It only retrieves rows that exist in both tables without considering matching columns.

Answer

- c) It automatically matches columns with the same name and eliminates duplicates.

- 28) What does the Len function in Visual Basic return? (2 marks)

- a) The number of spaces in a string
- b) The length of a string including spaces
- c) The number of characters excluding spaces
- d) The ASCII value of a string

Answer

- b) The length of a string including spaces

- 29) Determine whether the following statements about the key differences between DAO and ADO in Visual Basic are True or False:

(2 marks)

1. DAO is for desktop apps, while ADO is for web and distributed apps.
2. ADO is more flexible, accessing multiple data sources via OLE DB.

Answer

1. True
2. True

- 30) What is the role of the long-term scheduler? (2 marks)
- a) Assigns CPU to processes
 - b) Loads processes from secondary storage to memory
 - c) Manages swapping of processes
 - d) Handles I/O device allocation

Answer

- b) Loads processes from secondary storage to memory
- 31) Why is cache memory used between the CPU and main memory? (2 marks)
- a) To reduce access time for frequently used data
 - b) To increase memory size
 - c) To store permanent data
 - d) To improve memory allocation

Answer

- a) To reduce access time for frequently used data

- 32) Why is ArrayList preferred over arrays in Java? (2 marks)
- a) Arrays are faster but less flexible than ArrayLists.
 - b) ArrayLists resize dynamically and provide utility methods for manipulation.
 - c) Arrays allow insertion at any position.
 - d) ArrayLists are thread-safe by default.

Answer

- b) ArrayLists resize dynamically and provide utility methods for manipulation.

- 33) Why are vector graphics preferred for scalable images? (2 marks)
- a) They use less memory.
 - c) They do not lose quality when resized.
 - d) They are easier to edit.
 - b) They are pixel-based.

Answer

- c) They do not lose quality when resized.

- 34) Why is the close() function essential in file handling? (2 marks)

- a) It prevents memory leaks by releasing resources used by the file.
- b) It automatically deletes the file after use.
- c) It ensures all data is delete from the file before closing.
- d) It prevents other programs from accessing the file permanently.

Answer

- a) It prevents memory leaks by releasing resources used by the file.

35)What is the significance of the JSP engine Jasper in Tomcat? (2 marks)

- a) It parses and compiles Java code
- b) It handles servlet requests
- c) It converts JSP files into servlets
- d) It provides network connections

Answer

- c) It converts JSP files into servlets

36) Determine whether the following statements about the importance of user interface principles in project success are True or False: (2 marks)

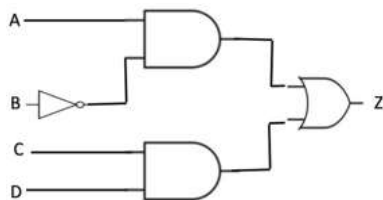
1. A well-designed user interface improves user experience and makes the application more intuitive.
2. Focusing on aesthetic design alone is enough for a successful user interface.

Answer

- a) True
- b) False

37) Design a circuit that implements the Boolean expression $Z = (A \cdot \neg B) + (C \cdot D)$. Include a labelled diagram. (5 marks)

Answer



38) Write a C++ program to print a multiplication table up to a given range using nested for loops. Your output must look like the following if the user input 10 as the range. (5 marks)

```

Enter the range for the multiplication table: 10
1x1=1 2x1=2 3x1=3 4x1=4 5x1=5 6x1=6 7x1=7 8x1=8 9x1=9 10x1=10
1x2=2 2x2=4 3x2=6 4x2=8 5x2=10 6x2=12 7x2=14 8x2=16 9x2=18 10x2=20
1x3=3 2x3=6 3x3=9 4x3=12 5x3=15 6x3=18 7x3=21 8x3=24 9x3=27 10x3=30
1x4=4 2x4=8 3x4=12 4x4=16 5x4=20 6x4=24 7x4=28 8x4=32 9x4=36 10x4=40
1x5=5 2x5=10 3x5=15 4x5=20 5x5=25 6x5=30 7x5=35 8x5=40 9x5=45 10x5=50
1x6=6 2x6=12 3x6=18 4x6=24 5x6=30 6x6=36 7x6=42 8x6=48 9x6=54 10x6=60
1x7=7 2x7=14 3x7=21 4x7=28 5x7=35 6x7=42 7x7=49 8x7=56 9x7=63 10x7=70
1x8=8 2x8=16 3x8=24 4x8=32 5x8=40 6x8=48 7x8=56 8x8=64 9x8=72 10x8=80
1x9=9 2x9=18 3x9=27 4x9=36 5x9=45 6x9=54 7x9=63 8x9=72 9x9=81 10x9=90
1x10=10 2x10=20 3x10=30 4x10=40 5x10=50 6x10=60 7x10=70 8x10=80 9x10=90 10x10=100

```

Answer

```

#include<iostream>
using namespace std;
int main() {
    int range;
    cout << "Enter the range for the multiplication table: ";
    cin >> range;
    for (int i = 1; i <= range; ++i) {
        for (int j = 1; j <= range; ++j) {
            cout << j << "x" << i << "=" << i * j << "\t";
        }
        cout << endl;
    }
    return 0;
}

```

39) Write a program that demonstrates encapsulation using a class. (5 marks)

Answer

```

#include <iostream>

using namespace std;

class BankAccount {
private:
    double balance;

public:
    void setBalance(double amount) {
        balance = amount;    }

    double getBalance() {
        return balance;    }
}

```

```
};  
  
int main() {  
    BankAccount account;  
  
    account.setBalance(1000.50);  
  
    cout << "Balance: " << account.getBalance() << endl;  
  
    return 0;}
```

40) Write a Java code snippet to demonstrate adding and retrieving elements from an ArrayList. (5 marks)

Answer

```
import java.util.*;  
public class ArrayListExample {  
    public static void main(String[] args) {  
        ArrayList<String> list = new ArrayList<>();  
        list.add("Apple");  
        list.add("Banana");  
        list.add("Cherry");  
        System.out.println("First element: " + list.get(0));  
        System.out.println("All elements: " + list);  
    }  
}
```

END

COMPUTER SCIENCE

Date: 14/ 06/2024

Period: 8:30 -11:30



END OF TERM III EXAMINATIONS QUESTION PAPER

GRADE: SENIOR FOUR (S4)

LEVEL: ADVANCED LEVEL

COMBINATIONS: -MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (MCE)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (MPC)

DURATION: 3HOURS

MARKS:

...../100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all 12 questions. (55marks)

Section B: Attempt 3 questions. (30marks)

Section C: Attempt 1 question. (15marks)

2)Use a blue or black ink pen only to write your answers.

Section A: Attempt all 12 questions from this section (55marks)

1. Why computer is said to be an electronic device? (3 marks)
2. List down the components of computer system. (4 marks)
3. Analyze why it is important to regularly blow out dust from a computer. (4 marks)
4. Explain why it is not advisable to take food substances and drinks in the computer lab. (3 marks)
5. Outline steps on how to access the disk management utility. (4 marks)
6. Workout the following expression (8 marks)
 - a) $1110_2 + 1111_2$ (2 marks)
 - b) $10101110_2 - 100110_2$ (2 marks)
 - c) $17.125_{10} = (\quad)_2$ (4 marks)
7. Is $F(X,Y,Z) = X + YZ$ equal to $F(X,Y,Z) = X + X + YZ$? (5 marks)
Explain your answer.
8. List down any three rules to be followed while naming a variable (6 marks)
9. State any five features of C++ programming language. (5 marks)
10. Explain the differences between pre-increment and post-increment operators in C++. When would you use each one? (4 marks)
11. Match the function with its description. (4 marks)
Functions:
 - a) Data Management
 - b) Security
 - c) Resource Management
 - d) User Interface
Description:
 1. Provides access to and controls hardware resources.
 2. Allows users to interact with the system.
 3. Protects the system from unauthorized access.
 4. Manages storage of files and data.

12. What are the utility functions of the following tags and attributes **(5 marks)**
- a. <body> </body>
 - b. <marquee> </marquee>
 - c. Valign
 - d. Href
 - e. <hr>

Section B: Attempt any (3) three questions from this Section (30marks)

13. By using Pseudocode method, write an algorithm to swap two numbers **(10 marks)**
14. In Fibonacci series, the next number is the sum of the previous two Fibonacci numbers as shown below: 0,1,1,2,3,5,8,13,21,
Based on the above scenario, write a C++ program that uses recursive function to output Fibonacci series from the first fifty natural numbers. **(10 marks)**
15. Using a sample program, demonstrate how you would use the get() function to read a string in as an array of characters. The output from the program should be displayed on the screen. **(10 marks)**
16. Differentiate between imperative, and functional programming paradigms. **(10 marks)**
17. Evaluate the difference between inline, internal, and external CSS. **(10 marks)**

Section C: Attempt any (1) one question from this Section (15marks)

18. Write an algorithm for the problems that allows the user to input three numbers and display the sum, the average, and their product. (Use for loop to input those numbers). (15 marks)

19. Create a C++ program that uses nested for loops and continue jump statement to format the following pattern: **(15 marks)**



20. The government of Rwanda is helping students wishing transfer from SCHOOL A to SCHOOL B. The students must send their request to NESA website using the form below. Write HTML code that displays this form. (15 marks)

BOARDING SCHOOL REQUEST FORM

STUDENT ID:

FIRST NAME:

LAST NAME:

AGE:

SEX:

PROVINCE:

DISTRICT:

SECTOR:

MARKS:

SCHOOL A:

SCHOOL B:

CLASS:

PHOTO:

COMPUTER SCIENCE

Date: 14/ 06/2024

Period: 8:30 -11:30



END OF TERM III EXAMINATIONS MARKING GUIDE

GRADE: SENIOR FOUR (S4)

LEVEL: ADVANCED LEVEL

COMBINATIONS: -MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (MCE)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (MPC)

DURATION: 3HOURS

...../100

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all 12 questions. (55marks)

Section B: Attempt 3 questions. (30marks)

Section C: Attempt 1 question. (15marks)

2)Use a blue or black ink pen only to write your answers.

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2024

Section A: Attempt all 12 questions from this section (55marks)

1. Why computer is said to be an electronic device? (3 marks)

Answer:

Computer is said to be an electronic device because it is made up of electronic components and uses electric energy (such electricity) to operate.

2. List down the components of computer system. (4 marks)

Answer:

Components of computer system are:

- Hardware
- Software
- Users
- Data

3. Analyze why it is important to regularly blow out dust from a computer. (4 marks)

Answer:

- To reduce wear and tear due to friction
- To reduce crashing of sensitive devices like the hard disk, fan, and RAM.
- Leaving the dust intact, can be a hazard to your computer.

4. Explain why it is not advisable to take food substances and drinks in the computer lab (3 marks)

Answer:

Drinks by drop on computer accessories causing short circuit or damage to electrical parts while solid food substances may cause clog moving parts or keys on the keyboard

5. Outline steps on how to access the disk management utility. (4 marks)

Answer:

- a) Right click This PC icon,
- b) Click on Manage to display Computer management window
- c) Click Disk management under storage
- d) Right click the drive you wish to manage on the right pane

6. Workout the following expression (8 marks)

a) $1110_2 + 1111_2$ (2 marks)

b) $10101110_2 - 100110_2$ (2 marks)

c) $17.125_{10} = (\quad)_2$ (4 marks)

Answer:

- a. 11101_2
- b. 10001000_2
- c. 10001.001_2

7. Is $F(X,Y,Z) = X + YZ$ equal to $F(X,Y,Z) = X + X + YZ$? **(5 marks)**
 Explain your answer.

Answer:

-

Yes.

- According to the identity law, $X+X = X$. Hence both statements can be written as: $X + YZ$.

8. List down any three rules to be followed while naming a variable **(6 marks)**

Answer:

- Choose meaningful variable names that tell the reader of the program what the variable represents.
- Each variable in the same algorithm should be identified using a unique name.
- By convention, variable names should begin with a letter of the alphabet but may be followed by numbers.
- Avoid using variable names that may conflict with reserved or keywords used in most programming languages.
- Variable names made up of two or more words should not have space in between the words, instead combine the two words or use an underscore.

9. State any five features of C++ programming language. **(5 marks)**

Answer:

- Portability
- Object-oriented programming
- Keywords
- Storage memory
- Identifiers
- Type checking

➤ Case sensitive

➤ Operators

10. Explain the differences between pre-increment and post-increment operators in C++. When would you use each one?

(4 marks)

Answer:

Pre-increment modifies the value and then uses it, while post-increment uses the current value first and then modifies it. Use pre-increment when the modified value is crucial, and post-increment when the original value matters.

11. Match the function with its description.

(4 marks)

Functions:

a) Data

Management b)

Security

c) Resource Management

d) User Interface

Description

:

1. Provides access to and controls hardware resources.
2. Allows users to interact with the system.
3. Protects the system from unauthorized access.
4. Manages storage of files and data.

Answer:

a)→ 4

b)→ 3

c)→ 1

d)→

2

12. What are the utility functions of the following tags and attributes

(5 marks)

a. <body> </body>

b. <marquee> </marquee>

c. Valign

d. Href

e. <hr>

Answer:

- a) **<body> </body>:** Encloses the body (text and tags) of the document
- b) **<marquee> </marquee>:** is used to move text in html
- c) **Valign:** The vertical alignment of the contents of cell
- d) **Href:** The URL of document to be linked to this one
- e) **<hr>:** horizontal rule line

Section B: Attempt any (3) three questions from this Section (30marks)

13. By using Pseudocode method, write an algorithm to swap two numbers
(10 marks)

Answer:

BEGIN

```

SET variable_A=0; Temporary=0; Variable_B=0
PRINT "Please enter two numbers"
READ Variable_A;
READ Variable_B Temporary
= Variable_A;
Variable_A=Variable_B;
Variable_B=Temporary;
PRINT Variable_A,Variable_B;

```

14. In Fibonacci series, the next number is the sum of the previous two Fibonacci numbers as shown below: 0,1,1,2,3,5,8,13,21,
Based on the above scenario, write a C++ program that uses recursive function to output Fibonacci series from the first fifty natural numbers.
(10 marks)

Answer:

```

#include<iostream>

using namespace std;

int fibonacci(int n){
    if((n==1) || (n==0)) {
        return(n);
    }
    else {
        return(fibonacci(n-1)+fibonacci(n-2));
    }
}

int main(){
    int n,i=0;
    cout<<"Enter number of terms for Fibonacci Series:";

    cin>>n;

    cout<<"The is the Fibonnaci Series";

```

```

while(i<n) {

cout<<" "<<fibonacci(i);

i++;

}

return 0;}

```

15. Using a sample program, demonstrate how you would use the `get()` function to read a string in as an array of characters. The output from the program should be displayed on the screen. (10 marks)

Answer:

```

#include <iostream>
using namespace std;
int main(){
    char Greeting[30];
    cout << "Greet someone:";
    cin.get(Greeting, 30); //enter 29 characters
    cout << "Greetings:"<<Greeting<< endl;
    return 0; }

```

16. Differentiate between imperative, and functional programming paradigms. (10 marks)

Answer:

Imperative programming focuses on giving instructions to the computer on how to perform a task. Programs written in an imperative style typically have a series of steps or commands that the computer follows in order to execute the program. Imperative programming languages such as C, Java, and Python are designed to support this paradigm.

Functional programming, on the other hand, focuses on using functions to perform tasks. Programs written in a functional style are made up of a series of functions that take input and produce output, without any side effects or state

changes. Functional programming languages such as Haskell, Lisp, and Clojure are designed to support this paradigm.

17. Evaluate the difference between inline, internal, and external CSS. (10 marks)

Answer:

Inline CSS: Styling applied directly to an HTML element using the style attribute.

```
<p style="color: blue;">This is a paragraph with inline CSS.</p>
```

Internal CSS: Styling defined within the <style> element in the <head> section of an HTML document.

```
<style>
  p {
    color: blue;
  }
</style>
```

External CSS: Styling defined in a separate CSS file and linked to the HTML document using the <link> element.

```
<link rel="stylesheet" href="styles.css">
```

Section C: Attempt any (1) one question from this Section (15marks)

18. Write an algorithm for the problems that allows the user to input three numbers and display the sum, the average, and their product. (Use for loop to input those numbers). (15 marks)

Answer:

```
Var number, I as integer
  Var sum, average, product as integer
  start
  Sum=0, product=1
  Write("enter numbers")
  for i=0 to 2 step 1
    read(number)
    sum=sum+number
    product=product*number
  end for
  average=sum/3
  write(sum)
  write(product)
  write(average)
end
```

19. Create a C++ program that uses nested for loops and continue jump statement to format the following pattern: **(15 marks)**



Answer:

```
#include <iostream>

using namespace std;

int main()
{
  int i,j,k;

  for(i=1; i<=7; i++)
  {
    if(i==3){
```

```

        continue;
    }
    for(j=6;j>=i;j--){
        cout<<" ";

    }
    for(k=1; k<=i; k++)
    {
        cout<<"* ";
    }
    cout<<"\n";
}
system("pause");
return 0;
}

```

20. The government of Rwanda is helping students wishing transfer from SCHOOL A to SCHOOL B. The students must send their request to NESA website using the form below. Write HTML code that displays this form. (15 marks)

BOARDING SCHOOL REQUEST FORM

STUDENT ID:	
FIRST NAME:	
LAST NAME:	
AGE:	
SEX:	v
PROVINCE:	
DISTRICT:	
SECTOR:	
MARKS:	
SCHOOL A:	
SCHOOL B:	
CLASS:	v
PHOTO:	Browse...
Send	

Answer:

```
<HTML> <b><caption>BOARDING SCHOOL REQUEST FORM
</caption><BR><BR></b> <form action="uploader.php" metho="post">
STUDENT ID:<input type ="text"><br> FIRST NAME:<input type="text"><br>
LAST NAME:<input type ="text"><br> AGE: <input type="text"><br> SEX:
<select><option>&nbsp;</option><option>male</option><option>female</opti
on><br ></select><br> PROVINCE:<input type="text"><br> DISTRICT:<input
type="text"><br> SECTOR:<input type="text"><br> MARKS:<input
type="text"><br> SCHOOL A: <input type="text"><br> SCHOOL B: <input
type="text"><br> CLASS:
<select><option>&nbsp;</option><option>s1</option><option>s2</option>
<option>s3</option><option>s4</option><option>s5</option>
<option>s6</option></select><br> PHOTO:<input name="uploadedfile"
type="file"/><br/> <input type ="submit" value="send"/> </FORM> </HTML>
```

COMPUTER SCIENCE

Date: 14/ 06/2024

Period: 8:30 -11:30



END OF TERM III EXAMINATIONS QUESTION PAPER

GRADE: SENIOR FIVE (S5)

LEVEL: ADVANCED LEVEL

COMBINATIONS: -MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (MCE)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (MPC)

DURATION: 3HOURS

MARKS:

...../100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all 12 questions. (55marks)

Section B: Attempt 3 questions. (30marks)

Section C: Attempt 1 question. (15marks)

2)Use a blue or black ink pen only to write your answers.

Section A: Attempt all 12 questions from this section**(55marks)**

1. Choose the correct answer, what is the most important safety precaution to take when cleaning a laptop? **(3 marks)**
 - a) Use harsh chemicals.
 - b) Power it off and unplug it.
 - c) Clean while it's running.
 - d) Don't worry about safety precautions.
2. Distinguish clearly between half-duplex and full-duplex in the context of data transmission. **(6 marks)**
3. Discuss any 3 Database models. **(6 marks)**
4. Define many to one relationship and give an example. **(5 marks)**
5. Explain how functional dependency differ from transitive dependency in database normalization **(5 marks)**
6. Differentiate between a class and an object in C++ with examples. **(5 marks)**
7. List any 5 basic Object Oriented Programming concepts. **(5 marks)**
8. Define the term debugging **(4 marks)**
9. What function is used to find the length of a string in Visual Basic? **(3 marks)**
 - a) Len()
 - b) Length()
 - c) StrLength()
 - d) StringLength()
10. Define method overloading and give an example. **(5 marks)**
11. When running a Java program with multiple classes, how does the Java Virtual Machine (JVM) identify the main class? **(4 marks)**
 - A) By searching for the class with the highest number of methods
 - B) By searching for the class with the shortest name

- C) By using the class specified with the -cp option
- D) By using the class specified with the public static void main(String[] args) method

12. What is the purpose of the "new" keyword in Java? **(4 marks)**
- A) It creates a new class definition.
 - B) It allocates memory for a new object.
 - C) It initializes instance variables within a class.
 - D) It defines a new method in a class.

Section B: Attempt any (3) three questions from this Section (30marks)

13. Write a C++ Program to find the Fibonacci series up to a given term. **(10 marks)**
14. Analyse the program given below and give its output appropriately. **(10 marks)**

```
#include <iostream>
int main() {
    int arr[5] = {10, 20, 30, 40, 50};

    int *ptr = &arr[4];

    std::cout << "Elements of the array accessed using pointer:\n";

    for (int i = 0; i < 5; ++i) {
        std::cout << "Element " << sizeof(arr)/sizeof(int)-i << ": " << *(ptr-i) << std::endl;
    }

    std::cout << "\nModified elements of the array using pointer:\n";

    for (int i = 4; i >=0; i--) {
        *(ptr-i) += 5;
        std::cout << "Element " << 5-i << ": " << arr[4-i] << std::endl;
    }
    return 0;
}
```

15. Consider the following declarations and answer the questions given below: **(10 marks)**

```
class living_being
{ char name[20];
protected:
    int jaws; public:
    void inputdata(char, int);
    void outputdata(); }
class animal: protected living_being {
    int tail;
protected:
    int legs;
public:
    void readdata(int, int);
    void writedata();
}; class cow: private animal { char horn_size;
public: void fetchdata(char); void displaydata(); };
```

- (i) Name the base class and derived class of the class animal.
- (ii) Name the data member(s) that can be accessed from function displaydata.
- (iii) Name the data member(s) that can be accessed by an object of cow class.
- (iv) Is the member function outputdata accessible to the objects of animal class.

16. Using label and textbox Write a VB program to calculate cube of an entered number. you should use vb6.0 or vb.net **(10 marks)**

17. Write Java program that uses an array to calculate the sum of the first

20 numbers.

(10 marks)

Section C: Attempt any (1) one question from this Section (15marks)

18. Write a program in Visual Basic that calculate a factorial of a number received through input box and displays the result in a message box. You should use vb6.0 or vb.net. **(15 marks)**
19. Write a Java program that reads data from a file named "source.txt" using FileInputStream and writes it to a file named "destination.txt" using FileOutputStream. **(15 marks)**
20. Observe the IP address 192.168.2.143/22 and respond to the questions below: **(15 marks)**
- What class does it belong
 - Write down the network id and host id of the above IP address
 - Which subnet mask does the network above have?
 - How many bits reserved to network?
 - How many hosts does the network above host?

COMPUTER SCIENCE

Date: 14/ 06/2024

Period: 8:30 -11:30



END OF TERM III EXAMINATIONS MARKING GUIDE

GRADE: SENIOR FIVE (S5)

LEVEL: ADVANCED LEVEL

COMBINATIONS: -MATHEMATICS -COMPUTER SCIENCE-
ECONOMICS (MCE)
- MATHEMATICS -PHYSICS-COMPUTER
SCIENCE (MPC)

DURATION: 3HOURS

MARKS:

...../100

CAMIS

..... /70

INSTRUCTIONS

1)This paper consists of three sections.

Section A: Attempt all 12 questions. (55marks)

Section B: Attempt 3 questions. (30marks)

Section C: Attempt 1 question. (15marks)

2)Use a blue or black ink pen only to write your answers.

Section A: Attempt all 12 questions from this section

(55marks)

1. Choose the correct answer, what is the most important safety precaution to take when cleaning a laptop? **(3 marks)**

- a) Use harsh chemicals.
- b) Power it off and unplug it.
- c) Clean while it's running.
- d) Don't worry about safety precautions.

Answer:

- b) Power it off and unplug it.

2. Distinguish clearly between half-duplex and full-duplex in the context of data transmission. **(6 marks)**

Answer:

Half-Duplex: the channel of communications allows alternating transmission in two directions, but not in both directions simultaneously

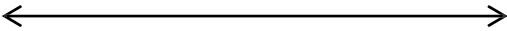
 *both*
direction on but not the same times



Example:

- Talkie- Walkie
- intercom,
- string phone

Full-Duplex: the communication between sender and receiver can occur simultaneously. or data flows in both directions at the same time.

 *both*
direction at the same times

Example:

- telephone conversation

3. Discuss any 3 Database models.

(6 marks)

Answer

The hierarchical model organizes its data using a tree structure. The root of the tree is the parent followed by child nodes. The hierarchical model

was widely used in the early days of computing, but it has since been replaced by more flexible models such as the relational model.

In the network model, entities are organized in a graph, in which some entities can be accessed through several paths. The network model is an extension of the hierarchical model, where each child node can have multiple parent nodes.

In the Relational Model, data is organized in two-dimensional tables called relations. The tables or relation are related to each other. The relational model is the most widely used database model.

4. Define many to one relationship and give an example. **(5 marks)**

Answer:

In a many-to-one relationship, multiple entities or records from one entity can be associated with a single entity or record from another entity.

For example, in a customer and orders scenario, a customer can place multiple orders, but each order is associated with only one customer. Therefore, the relationship between a customer and orders is a many-to- one relationship

5. Explain how functional dependency differ from transitive dependency in database normalization **(5 marks)**

Answer:

Functional Dependency:

Definition: Functional dependency describes a relationship between two attributes (or sets of attributes) in a relation such that one attribute's value determines another attribute's value.

Implication: Functional dependency is a fundamental concept used in normalization to identify and eliminate redundancy in a relation.

Transitive Dependency:

Definition: Transitive dependency occurs when a non-key attribute in a relation is functionally dependent on another non-key attribute rather than on a candidate key.

Implication: Transitive dependency violates the second normal form (2NF) in database normalization.

6. Differentiate between a class and an object in C++ with examples. **(5 marks)**

Answer:

A class is a template for creating objects, whereas an object is an instance of a class. For example, "Car" is a class, and "BMW" or "Toyota" are objects of the "Car" class.

7. List any 5 basic Object Oriented Programming concepts. **(5 marks)**

Answer:

Classes.

Objects

Encapsulation

Abstraction

Inheritance

Polymorphism

Interfaces

8. Define the term debugging **(4 marks)**

Answer:

Debugging is the process of identifying, isolating, and fixing errors or bugs in computer software or hardware. The goal of debugging is to ensure that the program runs as intended and produces the expected output without errors or unexpected behavior.

9. What function is used to find the length of a string in Visual Basic? **(3 marks)**

a) Len()

b) Length()

c) StrLength()

d) StringLength()

answer

a) Len()

10. Define method overloading and give an example. **(5 marks)**

Answer:

Method overloading is a feature in object-oriented programming where a class can have multiple methods with the same name but with different parameters or arguments. In method overloading, the methods must have the same name but can differ in the number, order, or type of parameters.

Here is an example of method overloading in Java:

```
public class Calculator {  
    public int add(int num1, int num2) {  
        return num1 + num2; }  
    public double add(double num1, double num2) {  
        return num1 + num2; }  
    public int add(int num1, int num2, int num3) {  
        return num1 + num2 + num3; } } }
```

11. When running a Java program with multiple classes, how does the Java Virtual Machine (JVM) identify the main class? **(4 marks)**

- A) By searching for the class with the highest number of methods
- B) By searching for the class with the shortest name
- C) By using the class specified with the -cp option
- D) By using the class specified with the public static void main(String[] args) method

Answer:

D) By using the class specified with the public static void main(String[] args) method

12. What is the purpose of the "new" keyword in Java? **(4 marks)**

- A) It creates a new class definition.
- B) It allocates memory for a new object.
- C) It initializes instance variables within a class.

D) It defines a new method in a class.

Answer

B) It allocates memory for a new object.

Section B: Attempt any (3) three questions from this Section (30marks)

13. Write a C++ Program to find the Fibonacci series up to a given term.

(10 marks)

Answer:

```
#include<iostream>
using namespace std;
main() {
    int n, c, first = 0, second = 1, next;
    cout << "Enter the number of terms of Fibonacci series you want" <<
endl;
    cin >> n;

    cout << "First " << n << " terms of Fibonacci series are :" << endl;

    for ( c = 0 ; c < n ; c++ ) {
        if ( c <= 1 )
            next = c;
        else
        {
            next = first + second;
            first = second;
            second = next;
        }
        cout << next << endl;
    }
    return 0;
}
```

14. Analyse the program given below and give its output appropriately.

(10 marks)

```
#include <iostream>
int main() {

int arr[5] = {10, 20, 30, 40, 50};
```

```

    int *ptr = &arr[4];
    std::cout << "Elements of the array accessed using pointer:\n";

    for (int i = 0; i < 5; ++i) {
    std::cout << "Element " << sizeof(arr)/sizeof(int)-i << ": " << *(ptr-i) << std::endl;

        }
    std::cout << "\nModified elements of the array using pointer:\n";

    for (int i = 4; i >=0; i--) {
        *(ptr-i) += 5;
    std::cout << "Element " << 5-i << ": " << arr[4-i] << std::endl;

        }
    return 0;
}

```

Answer

```

Elements of the array accessed using pointer:
Element 5: 50
Element 4: 40
Element 3: 30
Element 2: 20
Element 1: 10

Modified elements of the array using pointer:
Element 1: 15
Element 2: 25
Element 3: 35
Element 4: 45
Element 5: 55

```

15. Consider the following declarations and answer the questions given below: **(10 marks)**

```

class living_being {
    char name[20];
protected:
    int jaws; public:
    void inputdata(char, int);

```

```

void outputdata(); }

class animal: protected living_being {
int tail;
protected:
int legs;
public:
void readdata(int, int);
void writedata();
}; class cow: private animal { char horn_size;
public: void fetchdata(char); void displaydata(); };

```

- (i) Name the base class and derived class of the class animal.
- (ii) Name the data member(s) that can be accessed from function displaydata.
- (iii) Name the data member(s) that can be accessed by an object of cow class.
- (iv) Is the member function outputdata accessible to the objects of animal class.

Answer:

- (i) Base class: living_being Derived class: cow
- (ii) horn_size, legs, jaws
- (iii) fetchdata() and displaydata()
- (iv) No

16. Using label and textbox Write a VB program to calculate cube of an entered number. you should use vb6.0 or vb.net **(10 marks)**

Answer:

Vb6.0	Vb.net
<pre> Private Sub Command1_Click() Dim a As Integer Dim result As Integer a = Val(Text1.Text) result = a * a * a Label1.Caption = result End Sub </pre>	<pre> Public Class Form1 Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click Dim a As Integer Dim result As Integer a = Val(TextBox1.Text) result = a * a * a Label1.Text = result End Sub End Class </pre>

17. Write Java program that uses an array to calculate the sum of the first 20 numbers. **(10 marks)**

Answer:

```

public class SumOfFirst20Numbers {
public static void main(String[] args) {
int[] numbers = new int[20];
int sum = 0;
for (int i = 0; i < 20; i++) {
numbers[i] = i + 1; }
for (int i = 0; i < 20; i++) {
sum += numbers[i]; }
System.out.println("The sum of the first 20 numbers is: " + sum); } }

```

Section C: Attempt any (1) one question from this Section (15marks)

18. Write a program in Visual Basic that calculate a factorial of a number received through input box and displays the result in a message box. You should use vb6.0 or vb.net. **(15 marks)**

Answer:

<pre> Private Sub Form_Load() Dim n As Integer Dim fact As Long fact = 1 n = InputBox("enter a number here:", "Input Menu") For i = 1 To n fact = fact * i Next MsgBox (n & "!=" & fact) End Sub </pre>	<pre> Public Class Form1 Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load Dim n As Integer Dim fact As Long fact = 1 n = InputBox("enter a number here:", "Input Menu") For i = 1 To n fact = fact * i Next MsgBox(n & "!=" & fact) End Sub End Class </pre>
--	--

19. Write a Java program that reads data from a file named "source.txt" using FileInputStream and writes it to a file named "destination.txt" using FileOutputStream. **(15 marks)**

Answer:

import

```

java.io.*;
public class Filecopy {
    public static void main(String[] args) {
        // Step 1: Define the input and output filepaths
        String sourceFile = "source.txt";
        String destinationFile = "destination.txt";
        // Step 2: Create FileInputStream and FileOutputStream objects
        try (FileInputStream inputStream = new
FileInputStream(sourceFile);
            FileOutputStream outputStream = new
FileOutputStream(destinationFile)) {
            // Step 3: Read data from the source file and write it to the
destination file
            int data;
            while ((data = inputStream.read()) != -1) {
                outputStream.write(data);
            }
            // Step 4: Display a message indicating successful copy
            System.out.println("File copied successfully!");
        } catch (IOException e) {
            // Step 5: Handle any exceptions that may occur during file
operations
            e.printStackTrace();
        }
    }
}

```

20. Observe the IP address 192.168.2.143/22 and respond to the questions below: **(15 marks)**
- What class does it belong
 - Write down the network id and host id of the above IP address
 - Which subnet mask does the network above have?
 - How many bits reserved to network?
 - How many hosts does the network above host?

Answer:

- IT BELONG TO CLASS C
- NETWORK ID: 192.168.2 HOST ID 143
- SUBNET MASK 255.255.252.0
- 22 bits
- HOSTS: $2^{10}-2=1022$ HOSTS

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2023

SUBJECT: COMPUTER SCIENCE

CLASS: SENIOR FOUR

COMBINATIONS: MATHS-COMPUTER SCIENCE-ECONOMICS: MCE

MATHS-PHYSICS-COMPUTER SCIENCE: MPC

1. Define a supercomputer. (3marks)

Answer:

Supercomputers are the fastest, largest, most expensive and powerful computers available. They are able to perform many complex operations in a fraction of a second. Supercomputers are mainly used for scientific research, which requires enormous calculations. Some of the applications that justify use of supercomputers include aerodynamic design and simulation, petroleum research, defence and weapon analysis and telecommunications.

2. Discuss any six main factors of computer performance. (6marks)

Answer:

There are several factors that can influence the performance of a computer, including:

CPU: The Central Processing Unit (CPU) is the brain of the computer, and its speed and number of cores can have a significant impact on performance.

RAM: Random Access Memory (RAM) is a type of memory that the computer uses to store data and instructions that are currently being used. More RAM can improve the speed at which the computer can access and process data.

Storage: The type and speed of the storage device (e.g., hard disk drive, solid-state drive) can also affect performance, particularly in terms of how quickly data can be accessed and read.

Graphics card: For tasks that involve heavy graphics processing, such as gaming or video editing, a dedicated graphics card can significantly improve performance.

Operating System: The efficiency of the operating system, as well as the software and drivers that run on it, can also impact performance.

Power Supply: A stable and reliable power supply is important for maintaining consistent performance and avoiding system crashes.

Cooling: Heat can reduce the performance of a computer, so effective cooling systems such as fans or liquid cooling can help to maintain optimal performance.

3. Mention any four related health risks that can be caused by computer at workplace.

(4marks)

Answer:

Working at a computer can cause neck and shoulder pains, headache, eyestrain and overuse injuries of the arms and hands.

4. Identify the steps to uninstall a program in your computer window operating system.

(5 marks)

Answer:

The general steps that can be followed:

- Open the "Control Panel" on your computer. On Windows, you can access the Control Panel by clicking on the Windows Start button, and then selecting "Control Panel" from the list of options. On a Mac, click on the Apple menu, then select "System Preferences."
- Look for the "Programs" or "Programs and Features" section in the Control Panel. On Windows, it may be labelled as "Add or Remove Programs."
- Find the program you want to uninstall in the list of installed programs.
- Select the program and click on the "Uninstall" button. On Windows, this is usually located at the top of the list of installed programs.
- Follow the prompts to complete the uninstallation process. This may involve clicking through a few screens to confirm that you want to uninstall the program, and choosing any additional options for the uninstallation process.
- Once the program has been uninstalled, you may need to restart your computer to complete the process.

5. Draw NOR gate and its truth table.

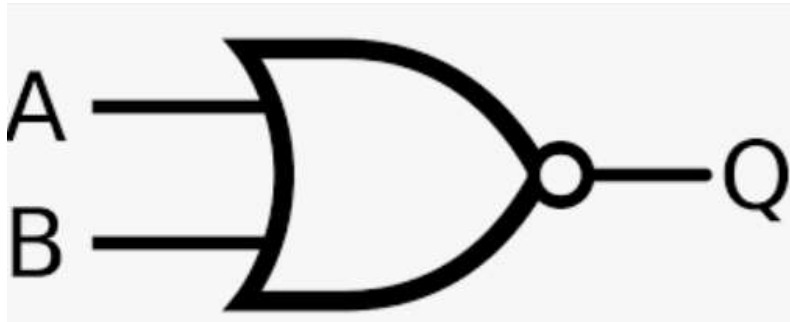
(6marks)

Answer:

TRUTH TABLE

A	B	ANORB
0	0	1
0	1	0
1	0	0
1	1	0

LOGIC GATE



6. What are the characteristics of a good algorithm? (5marks)

Answer:

Following are the characteristics of a good algorithm:

- a) It receives an input.
- b) It works on a set of inputs.
- c) The steps must be precisely defined.
- d) The result of each step should depend on the results of previous steps.
- e) It stops after a finite number of instructions.
- f) It produces the desired output.

7. Both conditional statement and iterative statement have a condition and a statement. How do they differ? (4marks)

Answer:

Conditional statement is executed only if the condition is true. Otherwise, nothing is done.

Iterative statement repeatedly evaluates a condition and executes a statement as long as the condition is true.

8. State any five factors that needed to be considered when declaring a one- dimensional array. (5marks)

Answer:

- Data type
- Array size
- Scope
- Initialization
- Array name

9. List three characteristics of procedural oriented programming. **(3marks)**

Answer:

3 characteristics of procedural oriented programming:

- Emphasis on procedures or functions
- Top-down approach
 - Lack of emphasis on data

10. State any six things that can cause your computer/Operating System to slow down?
(6marks)

Answer:

- You have too many start up programs
 - Your hard drive is failing
 - Your hard drive is 95% full
 - Your browser has too many add-ons
 - You're running too many programs at once
 - Too many browser tabs are open
 - You have a virus
 - You don't have enough memory
 - You need to restart your computer
 - You're running in low power mode
11. Which of the following is the correct syntax for referring the external style sheet?
(2marks)

- A. <style src = example.css>
- B. <style src = "example.css" >
- C. <stylesheet> example.css </stylesheet>
- D. <link rel="stylesheet" href="example.css">

Answer:

Answer is D

12. List any six examples of relational operators. **(6marks)**

Answer:

Relational Operators

- < : less than.
- <= : less than or equal to.
- > : greater than.
- >= : greater than or equal to.
- == : equal to.
- != : not equal to.

Section B

13. Elaborate responsibilities and role of technician in computer repairing.

(10marks)

Answer:

A computer repair technician works to ensure computers function correctly. Their responsibilities can vary depending on their work environment and role, but common duties include:

- Preparing new computer systems
- Installing software and computer programs
- Maintaining and troubleshooting existing software
- Updating and checking computer security programs
- Instruction customers on best security practices
- Analyzing computer function and repairing hardware or servers as needed
- Testing and evaluating new software programs for a company
- Assisting customers or employees with technical issues
- Answering questions and describing troubleshooting steps to customers
- Training customers or employees on computer software programs or general use
- Visiting homes or offices to repair computers
- Working remotely to solve problems through live chat and diagnostic testing.

14. Convert base number system

(10marks)

- a) $101111001102 = ?_{16}$
- b) $DDFF3416 = ?_{10}$
- c) $10028 = ?_{16}$
- d) $1100111001112 = ?_8$
- e) $100.1102 = ?_{10}$

Answer:

- a. 5E6
- b. 14548788
- c. 202
- d. 6347
- e. 4.75

15. Explain the tags that allow to:

(10marks)

- a) Create a paragraph
- b) An ordered List

c) An unordered List d)

Underline text

e) Boldface text

Answer:

a) To create a paragraph, you can use the HTML <p> tag.

b) To create an ordered list, you can use the HTML tag, and each item in the list should be wrapped in an tag.

c) To create an unordered list, you can use the HTML tag, and each item in the list should be wrapped in an tag.

d) The HTML <u> tag can be used to underline text.

e) The HTML or tag can be used to boldface text

16. Analyse the following statement and explain what will happen. when using for (i =

1; i >= 10; i++)

(10marks)

```
#include <iostream>
```

```
using namespace std; int
```

```
main()
```

```
{
```

```
int i,sum=0;
```

```
for (i = 1; i <= 10; i++)
```

```
{
```

```
sum=sum+i;
```

```
} cout <<sum << endl;
```

```
return 0;
```

```
}
```

Answer:

The statement for (i = 1; i >= 10; i++) sets the loop variable i to 1 initially and then checks if i is greater than or equal to 10. Since 1 is not greater than or equal to 10, the loop condition will be false and the loop will not be executed. Therefore, the code inside the loop will not be executed, and the value of sum will remain 0.

18. Write a C++ program to find the sum and average of one-dimensional integer array.
(10marks)

Answer:

```
#include<iostream>
using namespace std; int
main()
{
    int Arr[100],n,i,sum=0;
    cout<<"Enter number of elements you want to insert ";
    cin>>n;
    for(i=0;i<n;i++)
    {
        cout<<"Enter element "<<i+1<<":";
        cin>>Arr[i];
    }
    for(i=0;i<n;i++)
        sum+=Arr[i];
    cout<<"\nThe sum of Array is :"<<sum;
    cout<<"\nThe average of Array is :"<<sum/i;
    return 0;
}
```

Section c

18. Develop html code to produce the following table:

(15marks)

INNOVATION COMPANY					
Number of members					
branch A		branch B		Total members	
Boys	Girls	Boys	Girls	500	
150	70	120	140		

Answer:

```
<!DOCTYPE html>
<html>
<head>
<title>Table attributes </title>
</head>
<body>
<table border="4" cellpadding="3" cellspacing="3"
width="400"height="200">
<Caption><b> INNOVATION COMPANY</b></caption>
<tr>
<th colspan=6> Number of members</th>
</tr>
<tr>
<th colspan=2> branch A</th>
<th colspan=2> branch B</th>
<th colspan=2> Total members</th>
</tr>
<tr>
<th>Boys</th>
<th>Girls</th>
<th>Boys</th>
```

```

<th>Girls</th>
<th rowspan="2">500</th>
</tr>
<tr>
<td>150</td>
<td>70</td>
<td>120</td>
<td>140</td>
</tr>
</table>
</body>
</html>

```

19. If $A=4\pi r^2$, $V=\frac{4}{3}\pi r^3$, using manipulator functions `setw()` and `setprecision()`, Make a program used for calculating surface area and volume of a sphere so that the results are displayed correct to two decimal places. **(15marks)**

Answer:

```

#include <iostream>

#include <iomanip>

#include <cmath> using
namespace std; int main() {
const double PI = 3.14159;
double radius, area, volume;
cout << "Enter the radius of the sphere: ";
cin >> radius;
area = 4 * PI * pow(radius, 2);
volume = (4.0 / 3.0) * PI * pow(radius, 3);

```

```
cout << setw(20) << left << "Surface Area:" << fixed << setprecision(2) << right  
<< area << endl;
```

```
cout << setw(20) << left << "Volume:" << fixed << setprecision(2) << right <<  
volume << endl;
```

```
return 0; }
```

20. Write a function to calculate the factorial value of any integer as an argument. Call this function from main() and print the results in main().

(15marks)

Answer:

```
#include<iostream>  
using namespace std;  
int factorial(int a)  
{  
    int fact=1;  
    while(a>=1)  
    {  
        fact=fact*a;  
        a--;  
    }  
    return fact;  
}  
int main()  
{  
    int x,f;  
    cout<<"Enter number : ";  
    cin>>x;  
    f=factorial(x);  
    cout<<"The factorial is : "<<f;  
    return 0;  
}
```

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2023

SUBJECT: COMPUTER SCIENCE

CLASS: SENIOR FIVE

COMBINATIONS: MATHS-COMPUTER SCIENCE-ECONOMICS: MCE

MATHS-PHYSICS-COMPUTER SCIENCE: MPC

Section A: Attempt all 12 questions from this section (55marks)

1. Outline any six common ports we find on a laptop computer. (6marks)

Answer:

Six ports you might find on a laptop computer:

- USB (Universal Serial Bus) port
- HDMI (High-Definition Multimedia Interface) port
- Audio jack
- SD card slot
- VGA port
- Ethernet or RJ – 45 port

2. Searching is the process of finding the location of a given element in a set of elements. There are two simple approaches to searching using complex data structure. Summarize them. (4marks)

Answer:

Linear search: This method traverses a list sequentially to locate the search key.

Binary search: This method works on sorted lists by progressively making better guesses to find the location of a search key.

Linear search Algorithms

The **linear search** is also called sequential search. It is a simple method used for searching an array for a particular value. It works by comparing the value to be searched with every element of the array one by one in a sequence until a match is found.

Binary Search: You search an array by repeatedly dividing the search interval in half. While dividing you begin with an interval covering the whole array. If the value of the search key is less than the item in the middle of the interval, narrow the interval to the lower half. Otherwise narrow it to the upper half. Repeatedly check until the value is found or the interval is empty. This kind of search is also called the dichotomy method or bisection method.

3. Identify steps to locate Mac address of a computer. (4marks)

Answer

Locate Mac address of your computer

Step1: Click the Run button in the windows 10 Start Menu

Step 2: Type cmd in the Open prompt of the Run menu and click OK to launch a command prompt window

Step 3: Type ipconfig /all at the command prompt to check the network card settings.

Step4: MAC address is listed by ipconfig under Physical Address

4. Explain in brief the relational database model. **(5marks)**

Answer

The relational database model is a data storage and management system based on the concept of relations, or tables, that hold data in rows and columns. In a relational database, data is organized into tables with columns representing the attributes of the data, and rows representing individual records.

Each table has a unique identifier called a primary key, which is used to establish relationships between tables. Relationships can be one-to-one, one- to-many, or many-to-many.

The relational model also supports the use of SQL (Structured Query Language) to manipulate and query data. The advantages of the relational database model include flexibility, scalability, ease of use, and the ability to enforce data integrity through the use of constraints.

5. Let consider the database with table CUSTOMER identified by CustomerName, Address, TelephoneNumber, ItemOrdered, Quantity and Price. However, a customer may order several items and each customer in the database may order a different number of items. This situation makes it difficult to implement the data in a relational database, since we do not know how many order entries to allow.

Is this database structure complex? If yes, identify the anomalies it has and suggest the solution to the database designer to make it simple.

(5marks)

Answer:

Yes, this database structure is complex, because it results too much redundancy in the database's information. To avoid suffer from poor construction of this table, database designer is suggesting to divide into two tables: one for Customer and another one for Order, and make a relationship between those tables.

6. List any five symbols used in Entity Relationship Diagram. **(5marks)**

Answer:

The symbols commonly used in an Entity Relationship Diagram (ERD) are:

- Entities: Represented by rectangles

- Attributes: Represented by ovals
- Relationships: Represented by diamonds
- Cardinality: Represented by lines and arrows
- Primary key: Represented by an underline
- Foreign key: Represented by a dashed underline
- Weak entity: Represented by a double rectangle

7. Explain Object Oriented Programming concepts. **(5marks)**

Answer:

Object-Oriented Programming (OOP) is a programming paradigm that emphasizes the concept of "objects," which are instances of classes that encapsulate data and behaviour. OOP is widely used in software development because it makes it easier to manage complexity, reuse code, and maintain applications over time.

8. What do you understand by the term software deployment? **(4marks)**

Answer:

Software deployment refers to the process of making software applications or updates available for use on a computer system or device. It involves installing the software and configuring it to work properly within the target environment, such as a server, a desktop computer, or a mobile device.

9. Define repetition structure and give an example. **(4marks)**

Answer:

Repetition structures, also known as loops or iterative statements, are programming constructs that allow a sequence of statements to be executed repeatedly until a specific condition is met.

Common examples of repetition structures in programming include for loops, while loops, and do-while loops.

10. What is the output? **(5marks)**

```
class ArrayExample {
    public static void main(String[] args) {
        int[] age = {12, 4, 5, 2, 5};

        for (int i = 0; i < 5; ++i) {
            System.out.println("Element at index " + i + ": " + age[i]);
        }
    }
}
```

Answer:

Element at index 0: 12
Element at index 0: 4
Element at index 0: 5
Element at index 0: 2
Element at index 0: 5

11. List any four features of java programming language. **(4marks)**

Answer:

Key features of Java:

Platform Independence

Object-Oriented Programming

Multi-threading

Robustness.

Security

Performance

Scalability

Community Support

12. In Java, what are the four main tasks of JVM. **(4marks)**

Answer

The JVM performs the following main tasks: A) Loads code. B) Verifies code. C) Executes code. D) Provides runtime environment.

Section B: Attempt any (3) three questions from this Section (30marks)

13. Discuss different factors that must be considered before making a choice for the network Topology. (10marks)

Answer

The following are the key factors you should pay attention to when choosing a network topology.

□ **Budget:** On pricing matters, bus and ring topologies are quite cost-effective while star, mesh and tree and hybrid topologies are expensive.

□ **Hardware Resources:** Certain network topologies work best with certain hardware. And vice-versa. So before you make a decision on the topology to adopt, perform an inventory of your current hardware. Ease of Implementation: If you'll contract a third party to install and/or maintain your network, then the complexity of the network topology you choose is perhaps a non-issue.

□ **Size of Network:** How many devices are going to be on your network? How geographically dispersed are they? How far from the „center“ is the furthest device? Some topologies are inadequate or expensive when applied to large networks.

□ **Reliability:** When it comes to reliability, network topologies aren't created equal. If you are looking for high reliability because you are in an industry where even brief downtime and delays are frowned upon (e.g. banking), then network reliability is a fundamental consideration. Choose the topology that delivers the highest reliability.

□ **Future Expansion:** If you expect your organization to grow in size in the medium to long-term, Identify the topology that's easy to add new nodes to, without negatively affecting network performance.

14. What is the output of the program below, (10marks)

```
#include <iostream> using
namespace std; int main()
{
    int a = 11,*aptr; int b = 5,*bptr; int c; aptr=&a;  bptr=&b;
    cout << "*aptr*bptr is :" << *aptr*bptr << endl;
    cout << "a + b is :" << a+b << endl;
    cout << " *aptr-*bptr is :" << *aptr-*bptr << endl;
    cout << "a - b is :" << a-b << endl;
    cout << " *aptr**bptr is :" << *aptr**bptr << endl;
    cout << "a * b is :" << a*b << endl;
    cout << "a / b is :" << a/b << endl;
```

```
cout << "a % b is :" << a%b << endl;
cout << "a++ is :" << a++ << endl;

cout << "a-- is :" << a-- << endl; return 0; }
```

answer

Output:

```
*aptr+*bptr is :16 a +
b is :16

*aptr-*bptr is :6 a -
b is :6

*aptr**bptr is :55 a *
b is :55

a / b is :2 a
% b is :1

a++ is :11 a-
- is :12
```

15. Using class write a c++ program to find the largest number in two numbers entered by the user. **(10marks)**

Answer

```
#include<iostream>
using namespace std;
class biggest
{ private:
int a,b;
public:
void input();
void display();
};

void biggest::input()
```

```

{cout<<"Enter 2 numbers.:";
cin>>a>>b; }

void biggest::display()
{ if(a>b)
    cout<<"Biggest no.:"<<a;
    else
    cout<<"Biggest no.:"<<b; }

int main()
{
    biggest b;
    b.input();
    b.display();
    return 0; }

```

16. Distinguish between VB and C++.

(10marks)

Answer

VB (Visual Basic) and C++ are two programming languages used for developing software applications. While both are popular languages, they differ in several ways, including:

Syntax: VB uses a more natural language-like syntax, making it easier for beginners to learn and use. On the other hand, C++ has a more complex syntax, requiring more time and effort to master.

Object-Oriented Programming: Both VB and C++ support object-oriented programming (OOP), but C++ is considered to be more suited for OOP due to its flexibility and support for features such as multiple inheritance.

Platform: VB is primarily used for developing Windows applications, while C++ is a cross-platform language that can be used for developing applications for various operating systems.

Performance: C++ is generally faster and more efficient than VB due to its ability to directly access hardware resources and memory.

17. Make use of parameterized constructor to rewrite this program.

(10marks)

```

class employee {

```

```

int age;

float salary;

void display() { System.out.println(age+" "+salary);
}

public static void main(String args[]) {
    employee p1=new employee();  employee p2=new employee();  p1.display();
    p2.display(); }

```

Answer:

```

class employee {
    int age;
    float salary;
    employee(int a, float s) {
        age = a;
        salary = s;
    }
    void display() {
        System.out.println(age + " " + salary);
    }
    public static void main(String args[]) {
        employee p1 = new employee(25, 50000.0f);
        employee p2 = new employee(30, 60000.0f);
        p1.display();
        p2.display(); }}

```

Section C: Attempt any (1) one question from this Section (15marks)

18. Using destructor write a c++ program to calculate area of triangle.

(15marks)

Answer

```
#include <iostream>

#include <cmath> using
namespace std; class
Triangle {
private:
    double base, height, area;
public:
    Triangle(double b, double h) {
        base = b;
        height = h;
        area = 0.5 * base * height;
    }
    ~Triangle() {
        cout << "Triangle object destroyed." << endl;
    }
    void printArea() {
        cout << "Area of triangle: " << area << endl;
    }
};

int main() {
    double base, height;
    cout << "Enter the base of the triangle: ";
    cin >> base;
    cout << "Enter the height of the triangle: ";
    cin >> height;
```

```

Triangle t(base, height);

t.printArea();

return 0;}

```

19. By using While ...Wend loop Write a VB program that computes the sum of even numbers from 0 to 50. You may use VB6.0 or VB.net. **(15marks)**

Vb6.0	Vb.net
<pre> Private Sub Form_Load() Form1.Show Dim sum, i As Integer sum = 0 i = 0 While (i <= 50) If i Mod 2 = 0 Then sum = sum + i i = i + 2 End If Wend Label2.Caption = sum End Sub </pre>	<pre> Public Class Form1 Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load Dim sum, i As Integer sum = 0 i = 0 While (i <= 50) If i Mod 2 = 0 Then sum = sum + i i = i + 2 End If End While Label1.Text = sum End Sub End Class </pre>

20. Write a Java program that reads a number and check if it is palindrome or not. **(15marks)**

Answer

```

import java.util.Scanner;

public class Example24 {

    public static void main(String args[])
    {

        Scanner in = new Scanner(System.in);

```

```
System.out.print("Input a number: ");  
  
int n = in.nextInt();  
  
int sum = 0, r;  
  
    int temp = n;  
    while(n>0)  
    {  
        r = n % 10;  
        sum = (sum*10)+r;  
        n = n/10;  
    }  
    if(temp==sum)  
        System.out.println("It is a Palindrome number.");  
    else  
        System.out.println("Not a palindrome");  
    } }
```

Computer Science

Date: 22/06/2022
Period: 8H30'-11H30'



END OF TERM III EXAMINATIONS

LEVEL: Advanced Level, S4

COMBINATIONS MATHS-COMPUTER SCIENCE-ECONOMICS: MCE
 MATHS-PHYSICS-COMPUTER SCIENCE: MPC

DURATION: 3 Marks

MARKS:

..... / 100

INSTRUCTIONS

This paper consists of three sections: A, B and C.

Section A: Attempt all questions. (55 marks)

Section B: Attempt three questions. (30 marks)

Section C: Attempt any one question. (15 marks)

SECTION A: ATTEMPT ALL QUESTIONS.

(55 marks)

- 1) What are the different Operating Systems used in computer? (4 Marks)
- 2) Develop html code that displays the following page: (4 Marks)

I am coming from ~~United Kingdom~~ Rwanda

- 3) The HTML elements (Tags) can be classified as two types:

(1) Container elements, (2) Empty elements. Discuss on them and give an example. (4 Marks)

- 4) Distinguish between an algorithm and a process. (4 Marks)

- 5) Solve the following number system: (8 Marks)

- a) Convert $(1056)_{16}$ to octal number.
- b) Convert $(1001001100)_2$ to decimal number.
- c) Convert 10101_2 into an octal number.
- d) Convert hexadecimal 2C to decimal number.

- 6) Determine the role of Disk cleanup (3 Marks)

- 7) What is computer ergonomics? (3 Marks)

- 8) Discuss on functions of system software (4 Marks)

- 9) Define an Array? What are the types? (4 Marks)

- 10) What is the difference between “Run time error” and “Syntax error”? (4 Marks)

- 11) Explain the advantages of cascading style sheet. (4 Marks)

- 12) Explain how you would connect both data projector and monitor to a single computer. (4 Marks)

- 13) What is selection statement? write it's types? (5 Marks)

SECTION B: ATTEMPT THREE QUESTIONS.

(30 Marks)

- 14) Make comparison between fourth and fifth generation of computers. (10 Marks)
- 15) Draw flowchart to find area of a rectangle (10 Marks)
- 16) Write a C++ program that allows the user to enter marks for three subject. The program should calculate, then display the total and mean score of the three subjects. (10 Marks)
- 17) Write C++ programs to interchange the values of two variables. (10 Marks)
- 18) Write a function called zeroSmaller() that is passed two int arguments by reference then sets the smaller of the two numbers to 0. Write a main() program to exercise this function. (10 Marks)

SECTION C: ATTEMPT ANY ONE QUESTION.

(15 marks)

- 19) Explain the fundamental gates with expression and truth table. (15 Marks)
- 20) Write a C++ program to swap first and last element of an integer 1-d array. (15 Marks)

END !!!

Computer Science

MARKING GUIDE



END OF TERM III EXAMINATIONS

GRADE / LEVEL: Advanced Level S4

COMBINATIONS MATHS-COMPUTER SCIENCE-ECONOMICS: MCE
MATHS-PHYSICS-COMPUTER SCIENCE: MPC

DURATION: 3 Marks

MARKS: /100

INSTRUCTIONS

This paper consists of three sections: A, B and C.

Section A: Attempt all questions.	(55 marks)
Section B: Attempt three questions.	(30 marks)
Section C: Attempt any one question.	(15 marks)

**MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS
2022**

SECTION A: ATTEMPT ALL QUESTIONS.

(55 marks)

- 1) What are the different Operating Systems used in computer? (4 Marks)

Answer:

The different types of operating system used in the computer:

- (i) Single User and Single Task Operating Systems
 - (ii) Multi User Operating Systems
 - (iii) Multi Processing Operating Systems
 - (iv) Distributed Operating Systems
 - (v) Prominent Operating Systems
- 2) Develop html code that displays the following page: (4 Marks)

I am coming from ~~United Kingdom~~ Rwanda

Answer:

```
<html>  
<head>
```

```
<title> Inserting and Deleting text </title>
```

```
</head>
```

```
<body>
```

```
I am coming from <del> United Kingdom</del> <ins> Rwanda </ins>
```

```
</body>
```

```
</html>
```

- 3) The HTML elements (Tags) can be classified as two types: (1) Container elements, (2) Empty elements. Discuss on them and give an example.

(4 Marks)

Answer:

Container Elements: The tags which are required opening and closing is known as container elements or tags.

For example: <html>, <body>, <title>, <p> etc.,

Empty Elements: The tags which are required only opening tag is known as empty elements or tags.

For example:

- 4) Distinguish between an algorithm and a process. (4 Marks)

Answer:

Algorithm:

(i) An algorithm is a step-by-step sequence of statements to solve a problem.

(iii) As an algorithm is executed, a process evolves which solves the problem.

Process:

(i) An instruction describes an action.

(ii) When the instructions are executed, a process evolves which accomplishes

the intended task or solves the given problem.

- 5) Solve the following number system: (8 Marks)

a) Convert $(1056)_{16}$ to octal number.

b) Convert $(1001001100)_2$ to decimal

number. c) Convert 10101_2 into an octal number.

d) Convert hexadecimal 2C to decimal number.

Answer:

a) $(4182)_{10} = (10126)_8$ Therefore, $(1056)_{16} = (10126)_8$ /2 marks

b) $(1001001100)_2 = (588)_{10}$ /2 marks

c) $101012 = 258$ /2 marks

d) $2C_{16} = 44_{10}$ /2 marks

6) Determine the role of Disk cleanup (3 Marks)

Answer:

Disk clean-up utility can determine which files on hard drive may no longer be needed and delete those files. In addition to freeing up potentially significant amounts of hard drive space. Using this utility regularly can improve system performance.

7) What is computer ergonomics? (3 Marks)

Answer:

The term ergonomic refers to applied science of equipment design with the purpose of optimizing productivity while minimizing discomfort and fatigue. Good organic furniture and equipment helps in preventing health related risks such as arthritis, backache and fatigue.

8) Discuss on functions of system software (4 Marks)

Answer:

System software performs a variety of fundamental operations that avails computer

resources to the user. These functions include:

- Booting the computer and making sure that all the hardware elements are working properly.
- Performing operations such as retrieving, loading, executing and storing application programs.
- Storing and retrieving files.
- Performing a variety of system utility functions

9) Define an Array? What are the types? (4 Marks)

Answer:

"An array is a collection of variables of the same type that are referenced by a common name".

There are different types of arrays used in C++. They are:

- (i) One-dimensional arrays
 - (ii) Two-dimensional arrays
 - (iii) Multi-dimensional arrays
- 10) What is the difference between "Run time error" and "Syntax error"? (4 Marks)

Answer:

Syntax error

- (i) **Syntax** is a set of grammatical rules to construct a program.
- (ii) **Syntax errors** occur when grammatical rules of C++ are violated.
- (iii) **Example:** if a program tries to open a file which does not exist, it results in a run time error.

Run-time error

- (i) **A run time error** is that occurs during the execution of a program.
- (ii) It occurs because of some illegal operation that takes place.
- (iii) **Example:** if a program tries to open a file which does not exist, it results in a run time error.

- 11) Explain the advantages of cascading style sheet. (4 Marks)

Answer:

Maintainability: CSS are also defined and stored as separate files. So, the style and appearance of a web page can be dynamically changed and maintained with less effort.

Reusability: The styles defined in CSS can be reused in multiple HTML pages.

Easy to understand: The tags in web pages are well organized with style specifications and therefore it is easy to understand.

12) Explain how you would connect both data projector and monitor to a single computer. (4 Marks)

Answer:

You should have a computer with two VGA ports; Or one VGA port and a HDMI port so that you can connect one device to the VGA and the other to the HDMI. Alternatively, if the computer does not have two ports, you need to buy a VGA adapter that matches the ports on your devices.

After you connect the devices:

- (i) Click the Start button, choose Settings, and click the System icon.
- (ii) When the System page appears, click the words Advanced Display Settings in the screen's bottom-right corner.

13) What is selection statement? write it's types? (5 Marks)

Answer:

The selection statement means the statement(s) are executed depends upon a condition. If a condition is true, a true block (a set of statements) is executed otherwise a false block is executed. This statement is also called decision statement or selection statement because it helps in making decision about which set of statements are to be executed.

types are

- (i) if statement
- (ii) if-else statement
- (iii) nested if
- (iv) switch-case.

SECTION B: ATTEMPT THREE QUESTIONS.**(30 Marks)**

- 14) Make comparison between fourth and fifth generation of computers.
(10 Marks)

Answer:**The fourth generation computers have the following features:**

- a) These computers use LSI and VLSI technologies.
- b) Its sizes were reduced to desktop and laptop computer.
- c) These computers are highly reliable and accurate.
- d) They have a large memory and high functional speed.
- e) The operating speed is measured in beyond picoseconds and MIPS (Millions of instruction per second)
- f) Magnetic disk is the common source of external storage.
- g) Multiprocessing and multiprogramming OS (operating system) are used.
- h) 4GL are also used.

The examples are – IBM PC, Apple/Macintosh, 8086, 80286 SX, 80386 SX etc.

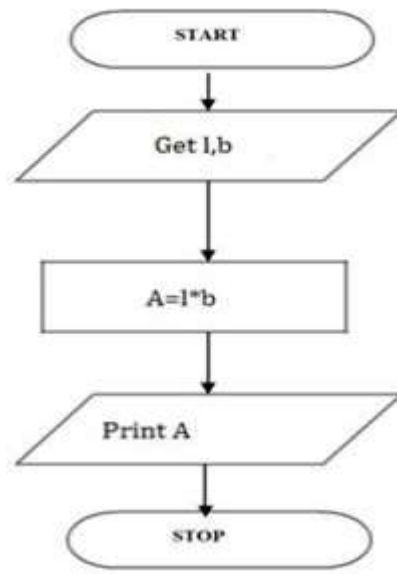
The computers having artificial intelligence (AI) and high processing capacity.

Fifth generation computers have the features:

- a) The speed will be extremely high in fifth generation computer.
- b) These computers will be using Ultra Large Scale Integration (ULSI) technology.
- c) The goal of fifth generation computers is to develop machines that will be able to think and take decisions.
- d) It can perform large number of parallel processing.
- e) Biochips and Gallium Arsenide (GaAS) will be used as memory devices.
- f) large uses of natural language processing and user friendly.
Able to make decisions like human beings.

- 15) Draw flowchart to find area of a rectangle
(10 Marks)

Answer:



- 16) Write a C++ program that allows the user to enter marks for three subject. The program should calculate, then display the total and mean score of the three subjects.
(10 Marks)

Answer:

```
#include <iostream>
using namespace std;
int main()
{
float a, b, c, sum; double mean;
cout << "Enter 1st number:";
cin >> a;
cout << "Enter 2nd number:";
cin >> b;
cout << "Enter 3rd number:";
cin >> c;
```

```

sum = a + b+c;
mean = sum/3;
cout << "Sum is:" << sum << endl;
cout << "Mean is:" << mean << endl;
return 0;
}

```

17) Write C++ programs to interchange the values of two variables. (10 Marks)

Answer:

```

#include<iostream>
using namespace std;
int main ()
{
int a, b, t;

cout<<"\n Enter First number";

cin>>a;

cout<<"\n Enter Second number";

cin>>b;

t=a;

a=b;

b=t;

cout<<a<<'\t'<<b;

return 0;

}

```

18) Write a function called zeroSmaller() that is passed two int arguments by reference then sets the smaller of the two numbers to 0. Write a main() program to exercise this function. (10 Marks)

Answer:

```
#include <iostream>
using namespace std;
int main()
{
    void zeroSmaller(int&, int&);
    int a=4, b=7, c=11, d=9;
    zeroSmaller(a, b);
    zeroSmaller(c, d);
    cout << "\na=" << a << " b=" << b
    << " c=" << c << " d=" << d;
    return 0;
}
void zeroSmaller(int& first, int& second)
{
    if( first < second )
        first = 0;
    else
        second = 0;
}
```

SECTION C: ATTEMPT ANY ONE QUESTION.**(15 marks)**

19) Explain the fundamental gates with expression and truth table.

(15 Marks)

Answer

A gate is basic electronic circuit, which operates on one or more signals to produce an output signal. There are three fundamental gates namely AND, OR and NOT.

AND Gate :

The AND gate can have two or more input signals and produce an output signal. In boolean algebra, a variable can take either of the values '0' or '1'. The logical symbol of the AND gate is

*Logic symbol of AND Gate*

In boolean algebra the multiplication sign stands for the AND operation. Therefore, the output of the AND gate is

 $C = A \cdot B$ or simply $C=AB$

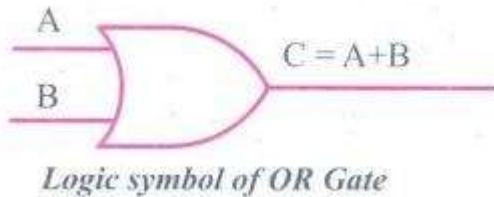
The truth table for AND Gate is

Input		Output
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

*The truth table for AND Gate**The truth table for AND Gate*

OR Gate :

The OR gate gets its name from the fact that it behaves like the logical inclusive "OR". The output is "true" if either or both of the inputs are "true". If both inputs are "false" then the output is "false". In other words the output will be 1 if and only if one or both inputs are 1; otherwise, the output is 0. The logical symbol of the OR gate is



Logic symbol of OR Gate

The OR gate output is

$$C = A \text{ OR } B$$

We use the + sign to denote the OR function.

Therefore,

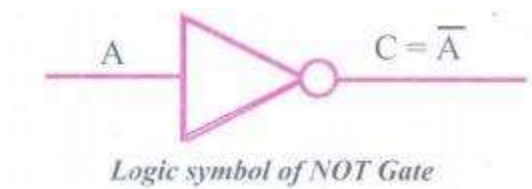
$$C = A + B$$

The truth table for OR gate is

Input		Output
A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

NOT Gate :

The NOT gate, called a logical inverter, has only one input. It reverses the logical state. In other words the output C is always the complement of the input. The logical symbol of the NOT gate is



Logic symbol of NOT Gate

The boolean function of NOT gate is

$$C = \text{NOT } A$$

In boolean algebra, the overbar stands for NOT operation. Therefore, $C = \bar{A}$
 The truth table for NOT gate is

Input	Output
A	C
1	0
0	1

20) Write a C++ program to swap first and last element of an integer 1-d array. (15 Marks)

Answer

```
#include<iostream>
using namespace std;
int main()
{
    int Arr[100],n,i,temp;
    cout<<"Enter number of elements you want to insert ";
    cin>>n;
    for(i=0;i<n;i++)
    {
        cout<<"Enter element "<<i+1<<":";
```

```
        cin>>Arr[i];
    }
    temp=Arr[0];
    Arr[0]=Arr[n-1];
    Arr[n-1]=temp;
    cout<<"\nArray after swapping"<<endl;
    for(i=0;i<n;i++)
        cout<<Arr[i]<<" ";
    return 0;
}
```

End !!!!

Computer Science

Date: 22/06/2022
Period: 8H30'-11H30'



END OF TERM III EXAMINATIONS

GRADE / LEVEL: Advanced Level, S5

COMBINATIONS MATHS-COMPUTER SCIENCE-ECONOMICS: MCE
 MATHS-PHYSICS-COMPUTER SCIENCE: MPC

DURATION: 3 Marks

MARKS: /100

INSTRUCTIONS

This paper consists of three sections: A, B and C.

Section A: Attempt all questions. (55 marks)

Section B: Attempt three questions. (30 marks)

Section C: Attempt any one question. (15 marks)

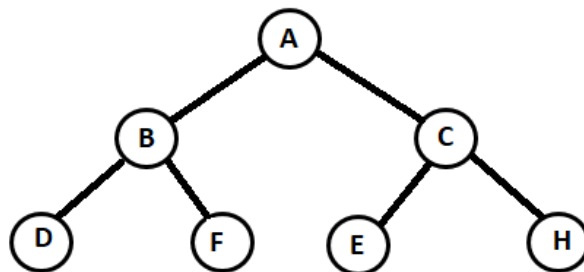
SECTION A: ATTEMPT ALL QUESTIONS.

(55 marks)

- 1) List any five examples of portable devices. (5 Marks)
- 2) List and define three standard streams in java. (6 Marks)
- 3) Define this() and super() in java. (5 Marks)
- 4) Prove the exact output produced by the following main() routine: (5 Marks)

```
public static void main(String[] args)
{ int x,y;
  x = 5; y = 1;
  while (x > 0)
  { x = x - 1; y = y * x;
    System.out.println(y); } }
```

- 5) Determine the order in which the elements would be accessed during Pre-Order, In Order and Post-Order traversals. (6 Marks)



- 6) Transmission media are broadly classified into two groups: Guided and Unguided media. Explain each group. (5 Marks)
- 7) A database contains a table with eight fields. The table currently has 100 records and each has data in each field. Answer the following question: (3 Marks)
 - a. How many entities are contained in the database?
 - b. How many attributes are contained in database?
 - c. How many individual pieces of data are in the database?

8) Suggest an output of the following program

(2 Marks)

```
#include<iostream>

using namespace std;

int main (){
int num, *ptr1, *ptr2;
ptr1=&num;
ptr2=ptr1+8;
cout<<"the difference is:"<<ptr2-ptr1<<endl;
return 0;
}
```

9) List any three advantages of inheritance

(3 Marks)

10) Answer TRUE or FALSE, the following sentences related to vb: (2 Marks)

- a) A project is an anything that is found on the computer specifically on the desktop.
- b) A class is collection of object with similar type

11) Write a class definition that creates a class called leverage with one private data member, crowbar, of type int and one public function whose declaration is void pry().

(4 Marks)

12) Write a Java program to print 'Hello' on screen and then print your name on a separate line. Using input from the user

(6 Marks)

13) You are browsing a website on the internet while the speed is okay and suddenly you find that the page is loading slowly. Investigate the reason behind slowness?

(3 Marks)

SECTION B: ATTEMPT THREE QUESTIONS.**(30 Marks)**

- 14) write a Java program to sort a numeric array and a string array.(10 Marks)
- 15) Compare computerized vs non-computerized database with at least 5 for each. (10 Marks)
- 16) Using a pointer write a C++ program which will ask the user to enter data and compute the area of a rectangle. Area = width*length. (10 Marks)
- 17) Build an application by using vb6.0 or vb.net to calculate the volume of the cylinder($V = \pi r^2 h$) (10 Marks)
- 18) a) Represent the employee relation in the conceptual model (4 Marks)

employees relation				
EmpId	FirstName	Surname	Address	Birthdate

- b) The following table describes information about employees, study it and answer the following questions. (6 Marks)

EmpId	EmpSname	EmpFname	EmpLocation	EmpJobCode
1011	RUKUNDO	Jean Claude	Kigali	12
1013	NYIRAMANA	Sylvane	Huye	19
1014	UWIMANA	Yvette	Muhanga	29
1021	KAGABO	Peter	Musanze	13

- What is the possible domain for field EmpId?
- How many records are shown?
- How many attributes are shown?

SECTION C: ATTEMPT ANY ONE QUESTION.

(15 marks)

19) Explain Elements of the Integrated Development Environment (IDE) in Visual basic (15 Marks)

20) Create a class student with the following specification (15 Marks)

Private members of class student:

admno integer

sname 20 character

eng, math, science float

total

float total() a function to calculate eng + math + science with float return type.

Public member function of class student

Takedata() Function to accept values for admno, sname, eng, science and invoke ctotat() to calculate total.

Showdata() Function to display all the data members on the screen.

End !!!!

Computer Science

MARKING GUIDE



END OF TERM III EXAMINATIONS

GRADE / LEVEL: Advanced Level S5

COMBINATIONS MATHS-COMPUTER SCIENCE-ECONOMICS: MCE
 MATHS-PHYSICS-COMPUTER SCIENCE: MPC

DURATION: 3 Marks

MARKS:

..... / 100

INSTRUCTIONS

This paper consists of three sections: A, B and C.

Section A: Attempt all questions.	(55 marks)
Section B: Attempt three questions.	(30 marks)
Section C: Attempt any one question.	(15 marks)

SECTION A: ATTEMPT ALL QUESTIONS.

(55 marks)

- 1) List any five examples of portable devices.

(5 Marks)

Answer:

Portable devices:

- a) Laptops
- b) Smartphones
- c) cell phones
- d) Personal Digital Assistants
- e) Tablets

- 2) List and define three standard streams in java.

(6 Marks)

Answer:

Java provides the following three standard streams:

- i. Standard Input: This is used to feed the data to user's program and usually a keyboard is used as standard input stream and represented as System.in.
- ii. Standard Output: This is used to output the data produced by the user's program and usually a computer screen is used to standard output stream and represented as System.out.
- iii. Standard Error: This is used to output the error data produced by the user's program and usually a computer screen is used to standard error stream and represented as System.err.

- 3) Define this() and super() in java.

(5 Marks)

Answer:

this() is used to access one constructor from another within the same class while **super()** is used to access superclass constructor. Either **this()** or **super()** exists it must be the first statement in the constructor.

- 4) Prove the exact output produced by the following main() routine:

(5 Marks)

```
public static void main(String[] args)
{ int x,y;
  x = 5; y = 1;
  while (x > 0)
  { x = x - 1; y = y * x;
    System.out.println(y); } }
```

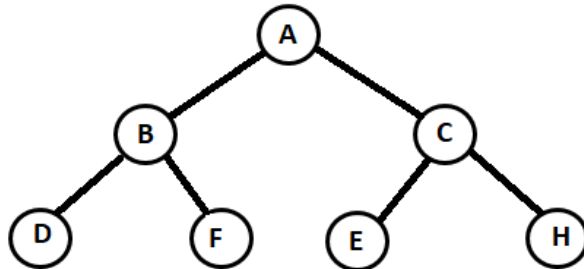
Answer:

Output:

4
12
24
24
0

- 5) Determine the order in which the elements would be accessed during Pre-Order, In Order and Post-Order traversals. (6 Marks)

Answer



Answer:

Pre-order: A B D F C E H

In-order: D B F A E C H

Post-order: D F B E H C A

- 6) Transmission media are broadly classified into two groups: Guided and Unguided media. Explain each group. (5 Marks)

Answer:

Wired or Guided Media: are the cables that are tangible or have physical existence and are limited by the physical geography. Example guided media are: Twisted pair cable, co-axial cable and fiber optical cable.

Wireless or Unguided Media are the ways of transmitting data without using any cables.

These media are not bounded by physical geography. Example of unguided media are: Radio wave, Infrared, Bluetooth.

- 7) A database contains a table with eight fields. The table currently has 100 records and each has data in each field. Answer the following question:

(3 Marks)

- How many entities are contained in the database?
- How many attributes are contained in database?
- How many individual pieces of data are in the database?

Answer:

a) one

entity b) 8
attributes

c) 800 piece of data Order, and make a relationship between those tables.

8) Suggest an output of the following program
(Marks)

(2)

```
#include<iostream>
using namespace std;
int main (){
int num, *ptr1,
*ptr2;
ptr1=&num;
ptr2=ptr1+8;
cout<<"the difference is:"<<ptr2-ptr1<<endl;
return 0;}
```

Answer:

The difference is 8

9) List any three advantages of inheritance
(Marks)

(3)

Answer:

1. Inheritance helps the code to be reused in many situations.(reusability)
2. It increases program structure which result in greater reliability.
3. Saves Time and Effort

10) Answer TRUE or FALSE, the following sentences related to vb: (2
Marks)

- a) A project is an anything that is found on the computer specifically on the desktop.
- b) A class is collection of object with similar type

Ans

wer

a)

False

b) True

11) Write a class definition that creates a class called leverage with one private data member,crowbar, of type int and one public function whose declaration is void pry(). (4

Marks)

Answer:

```
class leverage
{
private:
int crowbar;
public:
void pry();
};
```

- 12) Write a Java program to print 'Hello' on screen and then print your name on a separate line. Using input from the user (6 Marks)

Answer:

```
import
java.util.Scanner;
public class Main {
    public static void main(String[] args)
    {
        Scanner input = new Scanner (System.in);

        System.out.print("Input your first name:
"); String fname = input.next();
        System.out.print("Input your last name:
"); String lname = input.next();
        System.out.println();
        System.out.println("Hello \n"+fname+" "+lname);}}
```

- 13) You are browsing a website on the internet while the speed is okay and suddenly you find that the page is loading slowly. Investigate the reason behind slowness? (3 Marks)

Answer

Causes: bad made cables, malfunctioning of network devices, wireless interference and weather.

SECTION B: ATTEMPT THREE QUESTIONS.

(30 Marks)

- 14) write a Java program to sort a numeric array and a string array.(10 Marks)

Answer

```
package
arrayexercice;

import java.util.Arrays;
public class Exercise1 {
    public static void main(String[] args){
        int[] my_array1 = { 17, 20, 18, 14, 13,14, 24, 54, 72, 23, 56, 21, 57, 56};
        String[] my_array2 = { "Java", "Python", "PHP", "C#", "C Programming", "C++"
}; System.out.println("Original numeric array :
"+Arrays.toString(my_array1)); Arrays.sort(my_array1);
```

```

System.out.println("Sorted numeric array :
"+Arrays.toString(my_array1)); System.out.println("Original string array
: "+Arrays.toString(my_array2));

Arrays.sort(my_array2);

System.out.println("Sorted string array : "+Arrays.toString(my_array2));    } }

```

- 14) Compare computerized vs non-computerized database with at least 5 for each. (10 Marks)

Answer

Computerized

:

- Can hold a vast amount of data
- Very fast to find a specific record
- Can be used to analyze data
- Data can be sorted into ascending or descending order on multiple criteria
- Can easily update
- Records are stored safely; they are available when needed
- The database can be kept secure by use of passwords

Non computerized (paper based):

- Limited by physical storage space
- Difficult to search for a specific criteria
- Very difficult to analyze data
- Difficult to sort data on more than one criteria
- Records can be lost
- Difficult to make backup

- 16) Using a pointer write a C++ program which will ask the user to enter data

and compute the area of a rectangle. Area = width*length. (10 Marks)

Answer

```

#include
<iostream> using
namespace std; int
main()
{
int width,length,area,*p;

```

```

cout <<"\nenter the width";
cin>>width;

cout <<"\nenter the length";
cin>>length;

p = &width;

area = (*p) * length;

cout<<"The area of Rectangle is "<<area;


return 0;
}

```

- 17) Build an application by using vb6.0 or vb.net to calculate the volume of the cylinder($V = \pi r^2 h$) (10 Marks)

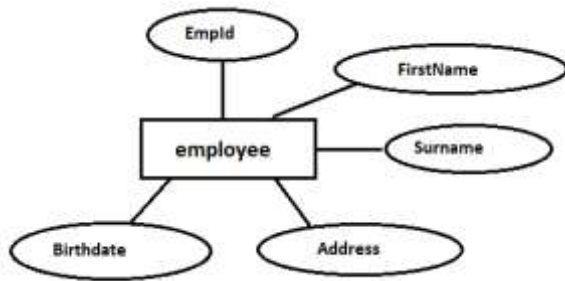
Answer:

Vb6.0	Vb14
<pre> Private Sub Command1_Click() Label1.Caption = 3.14 * Val(Text1.Text ^ 2) * Val(Text2.Text) End Sub Private Sub Form_Load() Text1.Text = "" Text2.Text = "" Label1.Caption = "" End Sub </pre>	<pre> Public Class Form1 Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click Label3.Text = 3.14 * Val(TextBox1.Text ^ 2) * Val(TextBox2.Text) End Sub Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load TextBox1.Clear() TextBox2.Clear() Label3.Text = "" End Sub End Class </pre>

- 18) a) Represent the employee relation in the conceptual model (4 Marks)

employees relation				
EmpId	FirstName	Surname	Address	Birthdate

Answer



b) The following table describes information about employees, study it and answer the following questions. (6 Marks)

EmpId	EmpSname	EmpFname	EmpLocation	EmpJobCode
1011	RUKUNDO	Jean Claude	Kigali	12
1013	NYIRAMANA	Sylvane	Huye	19
1014	UWIMANA	Yvette	Muhanga	29
1021	KAGABO	Peter	Musanze	13

- What is the possible domain for field EmpId?
- How many records are shown?
- How many attributes are shown?

Answer

- Domain of EmpId = {1011,1013,1014,1021}
- They are records 4 records
- They are 5 attributes

SECTION C: ATTEMPT ANY ONE QUESTION.

(15 marks)

19) Explain Elements of the Integrated Development Environment (IDE) in Visual basic (15 Marks)

Answer:

- a. Menu Bar: It contains a standard command like: File, Edit, View, Window, Help menus, and specific command such as: Project, Format, or Debug menus.
- b- Toolbar: it contains the most commonly used commands (button), if clicked an action represented by that button is carried out.
- c- ToolBox: it contains a collection of tools that are needed for project design
- d- Form Designer: it is a window for each form to customize the designed interface of the application. Using the form designer, the user can add controls, graphics, and text to create the desired form appearance.
- e- Properties Window: it is a List of properties settings for a selected form or a control. These properties are characteristics (such as size, visible, or color) of the selected object it provides an easy way to set properties.
- f- Project Explorer Window: it is a list of the forms and modules for the current projects. It is a hierarchical tree- branch structure, where the project at top of tree and other parts like forms ,modules) descend from this tree.
- g- Form Layout Window: The Form Layout window is a small screen. Which is used to reposition the form of the application so that it appears in proper place when project is run.

h- Code Editor Window: Code Editor Window is used to write a VB code for an application. For each form there is a separate code editor window. It is displayed when user clicks on form or object in form.

20) Create a class student with the following specification (15 Marks)

Private members of class student:

admno integer

sname 20 character

eng, math, science float

total

float total() a function to calculate eng + math + science with float return type.

Public member function of class student

Takedata() Function to accept values for admno, sname, eng, science and invoke cttotal() to calculate total.

Showdata() Function to display all the data members on the screen.

Answer:

```
#include<iostream>
using namespace std;
class student
{
private:
int admno;
char sname[20];
float eng,math,science;
float total;
float cttotal()
{
return eng+math+science;
}
public:
void Takedata()
{
cout<<"Enter admission number ";
cin>>admno;
cout<<"Enter student name " ;
cin>>sname;
cout<< "Enter marks in english, math, science ";
cin>>eng>>math>>science;
total=cttotal();
```

```

}
void Showdata()
{
    cout<<"Admission number "<<admno<<"\nStudent name
"<<sname<<"\nEnglish "  <<eng<<"\nMath "<<math<<"\nScience
"<<science<<"\nTotal "<<total;
}
};
int main ()
{
    student obj ;
    obj.Takedata();
    obj.Showdata();
    return 0;
}

```

End !!!!

COMPUTER SCIENCE

30/06/2021

8:30 am - 11:30am



SENIOR FOUR END OF YEAR EXAMINATIONS, 2020/2021

SUBJECT: COMPUTER SCIENCE

COMBINATIONS:

MATHEMATICS- PHYSICS- COMPUTER SCIENCE (MPC)

MATHEMATICS-COMPUTER SCIENCE –ECONOMICS (MCE)

Marks:

/100

DURATION: 3 HOURS

INSTRUCTIONS:

All questions are compulsory (100 marks)

Answer all questions (100 marks)

- 1) Explain why it is not advisable to eat or drink in a computer lab. **(3marks)**
- 2) Discuss Characteristics of Random Access Memory (RAM) **(6 marks)**
- 3) Differentiate between the Servers and workstations. **(4 marks)**
- 4) Which of the following is designed to control the operations of a computer? Explain it. **(5 marks)**
 - a) Application Software
 - b) System Software
 - c) Utility Software
 - d) User
- 5) State four factors you would consider before installing computer software whether an operating system or application software. **(4 marks)**
- 6) Convert 101102 into a decimal number. **(4 marks)**
- 7) Why the truth table of an AND gate has four possible combinations? **(3 marks)**
- 8) List six functions of the operating system. **(6 marks)**
- 9) Write a C++ program to assign a value of 100.235 to a double variable and then convert it to int. **(6 marks)**
- 10) List five rules you would follow to write code using XHTML **(5 marks)**
- 11) Draw the truth table of AND Gates **(4 marks)**
- 12) Write a C++ program to print the value of the address of the pointer to a variable whose value is input from user. **(10 marks)**
- 13) List 10 components inside computer case. **(10 marks)**
- 14) Write an algorithm which receive a width and height of a rectangle and display its area and perimeter **(10 marks)**
- 15) Explain major types of computer software, their uses, and examples. **(10 marks)**
- 16) Write pseudocode to compare three numbers X, Y and Z in algorithm. **(10 marks)**

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2021

Comprehensive Assessment S4

1) Explain why it is not advisable to eat or drink in a computer lab. **(3marks)**

Answer:

The food / drink may pour into sensitive computer parts causing clogging or rusting.

2) Discuss Characteristics of Random Access Memory (RAM) **(6 marks)**

Answer:

1. Data can be read (retrieved) and written (stored) in it.
 2. RAM is a temporary (volatile) storage because its content disappears when the computer is switched off.
 3. Its content is user defined i.e. the user dictates what is to be contained in the RAM.
- The two main types of RAM are:

3) Differentiate between the Servers and workstations. **(4 marks)**

Answer:

Server provides services to other computers on the network; Workstation used by the user, has limited processing power and requests services from the server.

4) Which of the following is designed to control the operations of a computer? Explain it. **(5 marks)**

- a) Application Software
- b) System Software
- c) Utility Software
- d) User

Answer:

b) System Software

Explanation:

System Software is basically classified into two: System and application. System Software is designed to control the operations and extend the processing capability of a computer system.

5) State four factors you would consider before installing computer software whether an operating system or application software. **(4 marks)**

Answer:

Memory (RAM) capacity.
Free hard disk space.
Processor type and speed.
Graphics display.

6) Convert 101102 into a decimal number.

(4 marks)

Answer:

The decimal equivalent number is given as:

$$\begin{aligned} &1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ &= 16 + 0 + 4 + 2 + 0 \\ &= 22_{10}. \end{aligned}$$

7) Why the truth table of an AND gate has four possible combinations?

(3 marks)

Answer:

-> It is because it has two entries making $2^2 = 4$ possible combinations

8) List six functions of the operating system.

(6 marks)

Answer:

cheduling
memory management
resource control and management
job scheduling
input/output management
job sequencing
security.

9) Write a C++ program to assign a value of 100.235 to a double variable and then convert it to int.

(6 marks)

Answer:

```
#include <iostream>
using namespace std;
int main()
{
    double x = 100.235;
    cout << (int)x << endl;
    return 0;}
```

10) List five rules you would follow to write code using XHTML

(5 marks)

Answer:

1. Write a DOCTYPE declaration at the start of the XHTML document.
2. All elements and attributes should be written in lower case e.g <body>.
3. Each opening tag must have an equivalent closing tag.
4. Nest all the tags properly.
5. Attribute values must be enclosed in quote marks e.g. <td rowspan = "3">.
6. Elements such as and <i> have been replaced by and respectively.

11) Draw the truth table of AND Gates

(4 marks)

Answer:

AND Gate		
A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

12) Write a C++ program to print the value of the address of the pointer to a variable whose value is input from user. (10 marks)

Answer:

```
#include<iostream>

using namespace std;

int main()
{
    int x, *y;
    cout << "Enter a number\n";
    cin >> x;
    y = &x;
    cout << "Value of the address of pointer of " << x << " is " << &y << "\n";
    return 0;
}
```

13) List 10 components inside computer case.

(10 marks)

Answer:

- Motherboard
- CPU/processor
- RAM (random access memory)
- Hard drive
- Power supply unit
- Expansion cards
- Video card
- Sound card
- Network card
- Fan

14) Write an algorithm which receive a width and height of a rectangle and display its area and perimeter
(10 marks)

Answer:

var a,b as double

Start

Write("enter a rectangle width")

Read(a)

Write("enter a rectangle height")

Read(b)

Write("the rectangle area is:",a*b)

Write("write the rectangle perimeter is:",(a+b)/2)

end

15) Explain major types of computer software, their uses, and examples.

Answer:

(10 marks)

System Software: System software coordinates the activities and functions of hardware and software, and it controls the operations of computer hardware and provides an environment or platform for all the other types of software to work in. It is the most basic type of software in any computer system, which is essential for other programs, applications and the whole computer system to function.

System software examples: Microsoft Windows XP, Mac O S, Linux, Windows Vista, Ubuntu, device drivers, etc.

Application Software: Application software is what helps the user perform the tasks of his/her choice. They are non-essential software which are installed and run, depending upon the requirements, in the environment provided by the system software.

Application software examples: MS Office, OpenOffice, Media Players, MS Access, educational software, media development software, Antivirus software, etc.

Programming Software: Programming software is used to write, test, debug and develop other software programs and applications. The various programming language editors such as Eclipse – a Java language editor – appear under this category. They are used for creating both the system as well as application software.

Programming software examples: Turbo C, Xilinx, Kiel, compilers, debuggers, Integrated Development Environment (IDE), etc.

16) Write pseudocode to compare three numbers X,Y, and Z in algorithm.

(10 marks)

Answer:

```
BEGIN
PRINT Enter x, y and z
READ x, y, z;
IF x < y Then
    IF x < z = Then;
    least = x
ELSE
    least = z
ELSE IF y < z Then
    least = y
ELSE
    least = z
ENDIF
PRINT least
END
```

COMPUTER SCIENCE

30/06/2021

8:30 am-11:30 am



SENIOR FIVE END OF YEAR EXAMINATIONS, 2020/2021

SUBJECT: COMPUTER SCIENCE

COMBINATIONS:

MATHEMATICS- PHYSICS- COMPUTER SCIENCE (MPC)

MATHEMATICS-COMPUTER SCIENCE –ECONOMICS (MCE)

DURATION: 3 HOURS

INSTRUCTIONS:

ANSWER ALL QUESTIONS (100 MARKS)

ANSWER ALL QUESTIONS (100 MARKS)

- 1) List any five internal components of a laptop **(5 marks)**
- 2) Give a clear difference between VGA and HDMI connectors. **(4 marks)**
- 3) Define two dimensional array and give their syntax for declaring it **(6 marks)**
- 4) Discuss the purpose of switching table in the switch? **(4 marks)**
- 5) Describer the 2 function of a modem **(4 marks)**
- 6) What is the purpose of managing information? **(4 marks)**
- 7) Explain three database design levels **(6 marks)**
- 8) By using a sketch, give types of Multiple Inheritance **(4 marks)**
- 9) What the following controls prefix stand for? Chk, fil,cbo,frm,txt. **(5 marks)**
- 10) What SQL command can be used to delete columns from a table? **(3 marks)**

A MODIFY TABLE TableName DROP COLUMN ColumnName
B MODIFY TABLE TableName DROP ColumnName
C ALTER TABLE TableName DROP COLUMN ColumnName
D ALTER TABLE TableName DROP ColumnName

- 11) Rewrite the following code using while loop: **(5 marks)**
- 12) Differentiate queue and stack and state their operations **(10 marks)**
- 13) Write an algorithm to solve quadratic equation. **(10 marks)**
- 14) Discuss advantages and disadvantage of using computer network **(10 marks)**
- 15) Write a java program that allows user to enter 2 inputs through the keyboard and calculate the sum, average and difference, use scanner **(10 marks)**
- 16) Write a vb program using do while...loop to display number from 0 to 100 **(10 marks)**

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2021

Comprehensive Assessment S5

- 1) List any five internal components of a laptop (5 marks)

Answer:

Ram, Cpu, Motherboard, Wireless card, Fan

- 2) Give a clear difference between VGA and HDMI connectors. (4 marks)

Answer:

HDMI (High-Definition Multimedia Interface) connector transports both sound and video data while VGA (Video Graphic Array) connector carries video data only. Note that both connectors transport any other type of data.

- 3) Define two dimensional array and give their syntax for declaring it (6 marks)

Answer:

A two dimensional array is a collection of a fixed number of elements (components) of the same Data type arranged in columns and rows making two dimensions.
The syntax for declaring a two dimensional array is:
Set array name= array [row size][column size] of data types

- 4) Discuss the purpose of switching table in the switch? (4 marks)

Answer:

The switch make a forwarding decision based on the destination host's MAC address. The switch essentially forwarded the frame from one port's ingress queue to another port's egress queue. To perform this forwarding decision, a switch consults its MAC address table.

- 5) Describer the 2 function of a modem (4 marks)

Answer:

The essential functions of a modem are:

Modulate: an analog carrier signal to carry digital information, it means to convert the analog signal to digital signal.

Demodulate: a similar signal so as to decode the digital information from the analog carrier signal and it means to convert the same signal back to the analog signal then transmitted through telephone line.

- 6) What is the purpose of managing information? (4 marks)

Answer:

The purpose of management information systems, including decision support systems, is to guide the decision-making process. Specifically, MIS combines people, computer hardware, software and communication devices to collect, store and convert raw data into useful information, which subsequently is communicated to the right people precisely when needed.

7) Explain three database design levels

(6 marks)

Answer:

Conceptual level: this level describes the logical structure of whole database by using different concepts like: Entity, occurrence, properties, cardinalities, relationships or associations.

Logical level: this describes how data are actually stored and how they can be accessed.

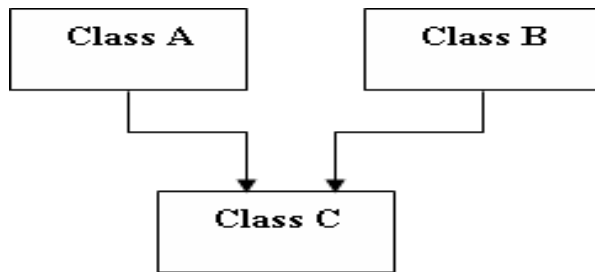
Physical level: in this level there exist a number of views which is defined apart of actual database by using the different DBMS.

8) By using a sketch, give types of Multiple Inheritance

(4 marks)

Answer:

Multiple Inheritance



9) What the following controls prefix stand for? Chk, fil,cbo,frm,txt. (5 marks)

Answer:

Chk: Check box

Fil: File list box

Cbo: combo box

Frm: form

Txt: text box.

10) What SQL command can be used to delete columns from a table?

(3 marks)

A MODIFY TABLE TableName DROP COLUMN ColumnName

B MODIFY TABLE TableName DROP ColumnName

C ALTER TABLE TableName DROP COLUMN ColumnName

D ALTER TABLE TableName DROP ColumnName

Answer:

C. ALTER TABLE TableName DROP COLUMN ColumnName

11) Rewrite the following code using while loop:

(5 marks)

```
inti,j;  
for(i=1,j=2;i<=6;i++,j+=2)  
System.out.println(i++);  
System.out.println("Finished!!!");
```

Answer:

```
inti=1,j=2;  
while(i<=6)  
{ System.out.println(i++);  
i++;  
j+=2;}  
System.out.println("Finished!!!");
```

12) Differentiate queue and stack and state their operations (10 marks)

Answer:

A queue is a linear list in which data can only be inserted at one end, called the REAR, and deleted from the other end, called the FRONT. These restrictions ensure that the data is processed through the queue in the order in which it is received. In an other words, a queue is a structure in which whatever goes fist comes out first (first in, first out (FIFO) structure).

Queue operations

There are two main operations related to queues.

Enqueue: the enqueue operation insert an items at the rear of the queue

Dequeue: the dequeue operation deletes the item at the front of the queue

Display: show elements in the array

A stack is a restricted linear list in which all additions and deletions are made at one end, the top.

The different operations performed on stacks are as follows:

Push: adds an element to the stack

Pop: removes an element from the stack

Peek: looks at the top element of the stack

13) Write an algorithm to solve quadratic equation.

(10 marks)

Answer:

Variables a, b, c, delta, x1, x2 : **numeric**

Begin

Write "Enter the coefficients of this polynomial:"

Read a ,b,c

Delta $\leftarrow b^2-4*a*c$

If delta>0 **Then**

x1 $\leftarrow b+\text{sqr}(\text{delta})/2*a$

x2 $\leftarrow b-\text{sqr}(\text{delta})/2*a$

Write " the equation has two distincts roots which are :",x1 , "nand ",x2

elseif delta=0 **Then**

x1 $\leftarrow$ x2 $\leftarrow b/2*a$;

Write"the equation has two equal roots : ",x1

else

Write " The equation has no real solutions. "

Endif

End

14) Discuss advantages and disadvantage of using computer network

(10 marks)

Answer:

Advantages of Computer Network

- It enhances communication and availability of information.
- It allows for more convenient resource sharing.
- It makes file sharing easier.
- It is highly flexible.
- It is an inexpensive system.
- It increases cost efficiency.
- It boosts storage capacity.

Disadvantages of Computer Network

- It lacks independence.
- It poses security difficulties.
- It lacks robustness.
- It allows for more presence of computer viruses and malware.
- Negative usage promotes negative acts.
- It requires an efficient handler.
- It requires an expensive set-up

15) Write a java program that allows user to enter 2 inputs through the keyboard and calculate the sum, average and difference, use scanner

(10 marks)

Answer:

```
import java.util.Scanner;
class AddNumbers
{
public static void main(String args[])
{
double x, y, sum, average,difference ;
System.out.println("Enter two numbers to calculate their sum");
Scanner in = new Scanner(System.in);
x = in.nextInt();
y = in.nextInt();
sum = x + y;
average= (x+y)/2;
difference=x-y;
System.out.println("Sum of the integers = " + sum);
System.out.println("average of the integers = " + average);
System.out.println("difference of the integers = " + difference);
}
}
```

16) Write a vb program using do while...loop to display number from
0 to 100 **(10 marks)**

Answer:

```
Private Sub Form_Load()  
Form1.Show  
Dim num As Integer  
num = 0  
Do While num <= 100  
num = num + 1  
print num  
Loop  
End Sub
```

COMPUTER SCIENCE

23/10/2019

2:00 pm - 5:00pm



SENIOR FOUR END OF YEAR EXAMINATIONS, 2019

SUBJECT: COMPUTER SCIENCE

COMBINATIONS:

MATHEMATICS- PHYSICS- COMPUTER SCIENCE (MPC)

MATHEMATICS-COMPUTER SCIENCE –ECONOMICS (MCE)

Marks:

/100

DURATION: 3 HOURS

INSTRUCTIONS:

All questions are compulsory **(100 marks)**

All questions are compulsory **(100 marks)**

- 1) Define a Computer (3 marks)

- 2) Discuss the Characteristics of Read Only Memory (ROM) (3 marks)

- 3) Outline the procedure you would follow to put out fire in a computer lab that may have been caused by electrical fault. (3 marks)
- 4) State three hardware requirements to be considered when installing application software. (6 marks)

- 5) Demonstrate and outline steps on how to access the disk management utility. (4 marks)

- 6) Define a Flowchart (3 marks)
- 7) Give an illustration of the following logic gates: NOT, AND (4 marks)

- 8) State three advantages of using functions. (6 marks)

- 9) Convert the following code segment into switch case construct. (4 marks)

```

int ch;
cin>>ch;
If(ch == 1)
{ cout<<" Laptop";
}
else If(ch == 2)
{
cout<<"Desktop ";
}
else if(ch == 3)
{
cout<<"Notebook";
}
else
{
cout<<"Invalid Choice";
}
}

```

- 10 a) Convert an octal number (345)₈ to a binary number (2 marks)

- b) Convert binary number (10011101)₂ to decimal (2 marks)

- 11) To reference storage of a variable in main memory, two operators, namely size of and address of (&) may be used, differentiate between the two operators. (4 marks)

12) Write the following acronyms in full: NTFS **(2 marks)**

13 Explain two component of CPU **(4 marks)**

14 a) Justify the reason why smartphones need an operating system. **(4 marks)**

b) List six functions of the operating system. **(6 marks)**

15 a) Write the following acronyms in full: **(2 marks)**

(i) HTML

(ii) XHTML

b) Write down html code that print the following **(8 marks)**

First Name:

Last Name:

Password:

16) Write program using an array to store 10 integers and show them. **(10 marks)**

17) Compare the advantages and disadvantages of high-level languages. **(10 marks)**

18) Write an algorithm which receives two numbers and informs the user whether their product is positive or negative. **(10 marks)**

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2019

SUBJECT: COMPUTER SCIENCE

CLASS: SENIOR FOUR

COMBINATIONS: MATHS-COMPUTER SCIENCE-ECONOMICS: MCE

MATHS-PHYSICS-COMPUTER SCIENCE: MPC

DURATION: 3HOURS

INSTRUCTIONS:

This paper consists of **two** sections: **A, B**.

Section A: Attempt all questions.

(50 marks)

1 Define a Computer

(3 marks)

Answer:

Computer: A computer is an electronic device capable of receiving raw facts (data) and performing a sequence of operations on the data based on special computer instructions (processing) to produce desired output (information).

2 Discuss the Characteristics of Read Only Memory (ROM)

(3 marks)

Answer:

1. One can only read its content but you cannot write on it unless it is a special type of ROM.
2. It is non-volatile i.e. its content is not lost when the computer is switched off.
3. Stores permanent or semipermanent instructions from the manufacturer called **firmware**. It can store semipermanent instructions because some variations of ROM chips can be programmed according to the user's specification.

3 Outline the procedure you would follow to put out fire in a computer lab that may have been caused by electrical fault.

(3 marks)

Answer:

Switch off power from main switch or circuit breaker; Use non-water non-powder based extinguisher to put out the fire.

4 State three hardware requirements to be considered when installing application software.

(6 marks)

Answer:

Memory requirements;
Processor speed;
Hard disk size

5 Demonstrate and outline steps on how to access the disk management utility.

(4 marks)

Answer:

1. Right click This PC icon/ Computer,
2. Click on Manage to display Computer management window
3. Click Disk management under storage
4. Right click the drive you wish to manage on the right pane

6 Define a Flowchart

(3 marks)

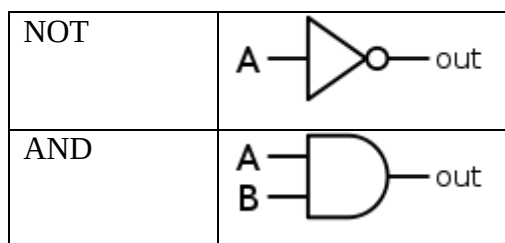
Answer:

Flowchart is a diagrammatic representation that illustrates the sequence of operations to be performed to get the solution of a problem. Is a graphical representation of an algorithm.

7 Give an illustration of the following logic gates: NOT, AND

(4 marks)

Answer:



8 State three advantages of using functions.

(6 marks)

Answer:

Advantages

- A modular program is easier to understand
- Easier to debug or modify a modular program
- Increases programmer productivity
- Libraries can be used in other programs

9 Convert the following code segment into switch case construct.

(4 marks)

```
int ch;
cin>>ch;
If(ch == 1)
{ cout<<" Laptop";
}
else If(ch == 2)
{
cout<<"Desktop ";
}
else if(ch == 3)
{
cout<<"Notebook";
}
else
```

```
{
cout<<"Invalid Choice";
}
```

Answer:

```
int ch;
cin>>ch;
switch(ch)
{
Case 1 : cout<<" Laptop"; break;
Case 2: cout<<"Desktop "; break;
Case 3: cout<<"Notebook";break;
Default : cout<<"Invalid Choice";
}
```

10 a. Convert an octal number (345)₈ to a binary number

(2 marks)

Answer:

Octal 3 is equivalent to 011
 Octal 4 is equivalent to 100
 Octal 5 is equivalent to 101
 Thus (345)₈ is equivalent to (011100101)₂

b. Convert binary number (10011101)₂ to decimal

(2 marks)

Answer:

Binary to decimal

Write each value of the binary number times two, from right to left, write the power of two starting from 0

$$1 \cdot 2^7 + 0 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = 157.$$

$$(10011101)_2 = (157)_{10}$$

11 To reference storage of a variable in main memory, two operators, namely size of and address of (&) may be used, differentiate between the two operators.

(4 marks)

Answer:

The sizeof() returns memory size allocated to a variable in bytes, while & returns actual address.

12 Write the following acronyms in full: NTFS

(2 marks)

Answer:

NTFS: New Technology File System

13 explain two component of CPU

(4 marks)

Answer:

The CPU is made up of three distinct components within it:

1. The Arithmetic Logic Unit (ALU): performs all arithmetic and logical operations.
2. Control Unit: interprets instructions and controls speed of execution using a clock.
3. Registers: special memories within the CPU for holding instructions and data.

Section B: Attempt all questions.**(50 marks)**

14 a. Justify the reason why smartphones need an operating system.

(4 marks)**Answer:**

Smartphones are special purpose computers. They have similar resources and are used to perform complex tasks similar computers hence the need for an operating system. They have a fast processor, large memory, many user applications, fast access to the internet etc.

b. List six functions of the operating system.

(6 marks)**Answer:**

- Scheduling
- memory management
- resource control and management
- job scheduling
- input/output management
- job sequencing
- security

15 a. Write the following acronyms in full:

(2 marks)

- (a) HTML
- (b) XHTML

Answer:

- (a) HTML - Hypertext Markup Language
- (b) XHTML - Extensible Hypertext Markup Language

b. Write down html code that print the following

(8 marks)

First Name:

Last Name:

Password:

Answer:

```
<!DOCTYPE html>
<html>
<head>
<form>
<body>
  First Name: <input type="text" name="firstname"/> <br/>
  Last Name: <input type="text" name="lastname"/><br/>
  <label for="password">Password: </label>
```

```

<input type="password" id="password" name="password"/><br/>
</body>
</form>
</head>

```

16 write program using an array to store 10 integers and show them.

(10 marks)

Answer:

```

#include<iostream.h>
main()
{
    int my_arr[5]; // name of array.
    cout<<"\nEnter values at: ";
    for(int i = 0 ; i < 10; i++)
    {
        cout<<"\n"<<i+1<<" :";
        cin>>my_arr[ i ]; //stores value at ith index.
    }
    for(int i = 0 ; i < 10; i++)
    {
        cout<<"\nNumber at "<<i+1<<" :"<<my_arr[ i ]; //show value at ith index.
    }
}

```

17 Compare the advantages and disadvantages of high-level languages.

(10 marks)

Answer:

Advantages and disadvantages of high-level languages:

Advantages

- High level languages are portable i.e. they are transferable from one computer to another.
- High level languages are user friendly and easy to use and learn.
- High level languages are more flexible, hence they enhance the creativity of the programmer and increase productivity in the workplace.
- A program in high level languages is easier to debug.

Disadvantages

- Their nature encourages use of many instructions in a word or statement hence the complexity of these instructions cause slower program processing.
- They have to be interpreted or compiled to binary form before the computer can execute them.
- They require large computer memory to run.

18 Write an algorithm which receives two numbers and informs the user whether their product is positive or negative.

(10 marks)

Answer:

SET a, b As Integer

start

WRITE("enter 1st number")

READ(a)

WRITE("enter 2nd number")

READ(b)

If $(a * b) > 0$ Then

WRITE("The product is positive")

Else If $(a * b) = 0$ Then

WRITE("the product is null")

Else

WRITE("the product is negative")

End If

end

COMPUTER SCIENCE

23/10/2019

2:00 pm - 5:00pm



SENIOR FIVE END OF YEAR EXAMINATIONS, 2019

SUBJECT: COMPUTER SCIENCE

COMBINATIONS:

MATHEMATICS- PHYSICS- COMPUTER SCIENCE (MPC)

MATHEMATICS-COMPUTER SCIENCE –ECONOMICS (MCE)

Marks:

/100

DURATION: 3 HOURS

INSTRUCTIONS:

All questions are compulsory (100 marks)

Attempt all questions. (100 marks)

1) Write an equivalent while loop for the following code: **(3 marks)**
Int sz=25;
for(int i=0,sum=0;i<sz;i++)
sum+=i;
System.out.println(sum);

2) List 5 access non modifiers in java **(5 marks)**

3) Explain the function of the following Stack Methods **(8 marks)**

Stack methods	Description
push ();	
pop ();	
peek ();	
boolean empty ();	

4) What are the disadvantage of a cell phone **(2 marks)**

5) Explain why Switch is much preferred than a Hub **(4 marks)**

6) Differentiate entity and relationship. **(3 marks)**

7) What are the purpose of managing information? **(2 marks)**

8) A School needs design a data base to store information about Teacher (identified by TeacherId, FirstName, Surname, Salary, Qualification, Address, Contact); Combination (identified by CombinationId, CombinationName), Student (identified by StudentId, FirstName, Surname, Address, Schoolfees, ContactNumber) and Subject (identified by subjectId, SubjectTitle).

As hired as data base design State database steps you will use to design that database.

(6 marks)

9) What the following controls prefix stand for? **(5 marks)**

Chk, fil, cbo, frm, txt

10) What do you understand by network subnet? **(3 marks)**

11) Write VB.6 codes that display your entered full name on the form when the application is loaded. **(4 marks)**

12) Predict the out put of the following program **(5 marks)**

```
class SampleSwitch {  
public static void main(String args[]) {
```

```

for(int i=0; i<6; i++)
switch(i) {
case 0:
System.out.println("i is zero.");
break;
case 1:
System.out.println("i is one.");
break;
case 2:
System.out.println("i is two.");
break;
case 3:
System.out.println("i is three.");
break;
default:
System.out.println("i is greater than 3.");
}
}
}

```

13) Write a java program where you declare the variables school, firstname, lastname, age, level and combination. This program should display like this: Gs Kigali, Ange Batoni is 19 years old and in senior 4 of LKK **(10 marks)**

14) Using Visual Basic, write a program that prompts a user to enter a student's score in Mathematics. If the score is above 45%, the program should display "Pass" otherwise it should display "Fail" **(10 marks)**

15) Write a class in C++ called person which has 2 functions: getdata() and putdata(). The getdata() prompts the user to enter his or her first name name, last name and age, the putdata() displays on the screen the user first name last, name and age. **(10 marks)**

17) Discuss advantages and disadvantage of using computer network in your school laboratory **(10 marks)**

18) Explain how you can care your laptop. **(10 marks)**

MARKING GUIDE FOR COMPREHENSIVE ASSESSMENT EXAMINATIONS 2019

SUBJECT: COMPUTER SCIENCE

CLASS: SENIOR FIVE

COMBINATIONS: MATHS-COMPUTER SCIENCE-ECONOMICS: MCE

MATHS-PHYSICS-COMPUTER SCIENCE: MPC

DURATION: 3HOURS

INSTRUCTIONS:

This paper consists of **two** sections: **A, B.**

Section A: Attempt all questions.

(50 marks)

1 Write an equivalent while loop for the following code:

(3 marks)

```
Int sz=25;
for(int i=0,sum=0;i<sz;i++)
sum+=i;
System.out.println(sum);
```

Answer:

```
int sz=25;
int i=0,sum=0;
while(i<sz)
{
sum+=i;
i++;
}
System.out.println(sum);
```

2 List 5 access non modifiers in java

(5 marks)

Answer:

Non-access modifiers do not change the accessibility of variables and methods, but they do provide them special properties. Non-access modifiers are of 5 types,

- Final
- Static
- Transient
- Synchronized
- Volatile

3 Explain the function of the following Stack Methods

(8 marks)

Answer:

Stack methods

Description

push ();

add/inserts new element/object into stack

pop ();

removes and return top of the stack

peek ();

returns top of the stack (just returns but doesn't remove unlike pop () operation)

boolean empty ();

check whether invoking stack is empty or not returns true, if stack is empty; otherwise, false

4 What are the disadvantage of a cell phone

(2 marks)

Answer:

One major disadvantage of a cell phone is its negative health effects (sleeplessness, eye cancer, infertility, cardiac problems, and chronic fatigue).

5 Explain why Switch is much preferred than a HUB

(4 marks)

Answer:

A **switch** is effectively a higher-performance alternative to a hub. For a network with more computers switch is preferable because it's built in MAC table. Technically speaking, **hubs** operate using a broadcast model where message is received on one port is generated and send to all ports of the hub and switches operate using a virtual circuit where received on one port is directly sent to its destination. Therefore, switch is much intelligent that a hub.

6 Differentiate entity and relationship.

(3

marks)

Answer:

Entity, defined as table/object that holds specific information (data). Student and Teacher are example of entities

Relationships, defined as the associations or interactions between entities. For example Student is taught by Teacher

7 What are the purpose of managing information?

(2 marks)

Answer:

The purpose of management information systems, including **decision support systems**, is to guide the decision-making process. Specifically, MIS combines people, computer hardware, software and communication devices to collect, store and convert raw data into useful information, which subsequently is Communicated to the right people precisely when needed.

8 A School needs design a data base to store information about Teacher (identified by TeacherId, FirstName, Surname, Salary, Qualification, Address, Contact); Combination (identified by

CombinationId, CombinationName), Student (identified by StudentId, FirstName, Surname, Address, Schoolfees, ContactNumber) and Subject (identified by subjectId, SubjectTitle).

As hired as data base design State database steps you will use to design that database.

(6 marks)

State database design steps.

Answer:

- Investigate the information.
- Identify the objects.
- Model the ob jects.
- Identify the types of information for each object.
- Identify the relationships between objects.
- Database optimisation through normalisation
- Data entry and manipulation

9 What the following controls prefix stand for? Chk, fil, cbo, frm, txt

(5 marks)

Answer:

- Chk: Check box
- Fil: File list box
- Cbo: combo box
- Frm: form
- Txt: text box

10 What do you understand by network subnet?

(3 marks)

Answer:

Network Subnet is logical division of the network and it is the strategy used to partition a single physical network into more than one smaller logical sub network.

11 Writes VB.6 codes that display your entered full name on the form when the application is loaded.

(4 marks)

Answer:

```
Private sub command1_click ()  
Dim name as string  
Name=Inputbox ("enter your full name")  
Print "my full name is"; name  
End sub  
Or  
Private sub form1_load ()  
Form1.show  
Print "RUBANGURA Simoni"
```

12 Predict the out put of the following program
class SampleSwitch {

(5 marks)

```

public static void main(String args[]) {
    for(int i=0; i<6; i++)
    switch(i) {
        case 0:
            System.out.println("i is zero.");
            break;
        case 1:
            System.out.println("i is one.");
            break;
        case 2:
            System.out.println("i is two.");
            break;
        case 3:
            System.out.println("i is three.");
            break;
        default:
            System.out.println("i is greater than 3.");
    }
}
}
}

```

Answer:

The output produced by this program is shown here:

```

i is zero.
i is one.
i is two.
i is three.
i is greater than 3.
i is greater than 3.

```

Section B: Attempt all questions.

(50 marks)

13 Write a java program where you declare the variables school, firstname, lastname, age, level and combination. This program should display like this: Gs Kigali, Ange Batoni is 19 years old and in senior 4 of LKK

(10 marks)

Answer:

```

package education;
public class education{
    public static void main(String[] args) {
        String school="GS Kigali", firstname = "Ange", lastname="Batoni";
        String level="senior4";
        String combination="LKK";
        int age =19;
        System.out.println(school+" "+firstname+" "+lastname+" "+"is"+" "+age+"years old and
in"+level+"
"+"of"+" "+combination);
    }
}

```

14 Using Visual Basic, write a program that prompts a user to enter a student's score in Mathematics. If the score is above 45%, the program should display "Pass" otherwise it should display "Fail" **(10 marks)**

Answer:

```
Private Sub Command1_Click()  
Dim mark As Single  
mark = Val(Text1.Text)  
If (mark > 45 And mark <= 100) Then  
Label3.Caption = "Pass"  
ElseIf (mark <= 45 And mark >= 0)  
Then  
Label3.Caption = "Fail"  
Else  
Label3.Caption = "Invalid"  
End If  
End Sub
```

15 Write a class in C++ called person which has 2 functions: getdata() and putdata(). The getdata() prompts the user to enter his or her first name name, last name and age, the putdata() displays on the screen the user first name last, name and age. **(10 marks)**

Answer:

```
#include<iostream>  
using namespace std;  
class person  
{  
private:  
char  
fname[20],lname[20];  
int age;  
public:  
void getdata()  
{  
cout<<"enter the first name, last  
name and age";  
cin>>fname>>lname>>age;  
}  
void putdata()  
{  
cout<<"First  
name: "<<fname<<endl;  
cout<<"Last  
name: "<<lname<<endl;  
cout<<"age:  
<<age<<endl;  
}  
};  
intmain()  
{
```

```
person a1;  
a1.getdata();  
a1.putdata();  
return 0;  
}
```

17 Discuss advantages and disadvantage of using computer network in your school laboratory
(10 marks)

Answer:

- Advantages of Computer Network
- It enhances communication and availability of information.
- It allows for more convenient resource sharing.
- It makes file sharing easier.
- It is highly flexible.
- It is an inexpensive system.
- It increases cost efficiency.
- It boosts storage capacity.

Disadvantages of Computer Network

- It lacks independence.
- It poses security difficulties.
- It lacks robustness.
- It allows for more presence of computer viruses and malware.
- Negative usage promotes negative acts.
- It requires an efficient handler.
- It requires an expensive set-up.

18 Explain how you can care your laptop.
(10 marks)

Answer:

- Always place your laptop on a level surface.
- Wash your hands before using the computer.
- Fluids are the number one enemy of electronic items so it is best not to eat or drink while working on the laptop.
- If liquid spills on the keyboard, turn the keyboard upside down to drain for at least 12 hours before powering on again.
- Never put personal data or programs that can't be easily recovered on your laptop.
- Your laptop may, someday, unexpectedly require service. Your personal data will not be protected if this happens and you could lose your data.
- Don't save passwords on your laptop.
- Use a password that is easy for you to remember (but not easy for someone else to guess)